MINISTRY OF EDUCATION AND TRAINING
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GRADUATION THESIS

Major: Information and Communication Technology

A NETWORK QUALITY OF SERVICE (QoS) SCHEDULING SYSTEM ON THE ANDROID PLATFORM

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Ha Noi, June 2026

DECLARATION OF ORIGINALITY

I hereby declare that this thesis is the result of my own original research, conducted under the academic supervision of Le Huy Quang.

All data, experimental results, and findings presented in this thesis are truthful and have not been previously published in any other work. All referenced materials have been fully and properly cited.

I accept full responsibility for the content and integrity of this thesis.

Ha Noi, 3 June 2026

Student

Pham Hieu Minh

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Finally, I wish to extend my heartfelt thanks to my family and friends, who have continuously encouraged and supported me during this time.

Despite my best efforts, due to limitations in knowledge and time, this thesis may contain certain shortcomings. I sincerely welcome any constructive feedback from the faculty and peers to further improve this work.

Ha Noi, June 2026

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ABSTRACT

As the number of mobile applications continues to grow and network usage demands diversify, controlling Quality of Service (QoS) on mobile devices has become a critical technical challenge. Existing solutions on Android either require Root privileges — impractical for most users — or only provide basic firewall functionality without priority-based bandwidth scheduling capabilities.

This thesis presents the design and implementation of **QoS Scheduler** — an Android application capable of intervening at the network layer (Layer 3) to perform per-application bandwidth scheduling **without requiring Root access**. The system utilizes the TUN virtual network interface through the Android VpnService API to intercept, analyze, and schedule all IP traffic on the device.

The main contributions of this thesis include:

- A Data Plane module capable of bit-level IPv4/IPv6 and TCP/UDP header parsing, including IPv6 Extension Header Chaining support.
- Implementation of the **Token Bucket** algorithm for rate limiting and **Weighted Fair Queuing (WFQ)** for priority-weighted bandwidth allocation (HIGH/MEDIUM/LOW).
- A UID Resolution mechanism based on 5-tuple information for per-application traffic classification.
- A stable TCP/UDP Relay system compatible with multiple Android versions, including custom ROMs such as HyperOS/MIUI.

Experimental results demonstrate the system's ability to accurately limit bandwidth, allocate network resources by weight with less than 5% deviation, and mitigate Bufferbloat for high-priority applications.

Keywords: Quality of Service, QoS, Android, VPN, Token Bucket, Weighted Fair Queuing, Packet Analysis, Bandwidth Scheduling.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ACK	Acknowledgement
API	Application Programming Interface
DPI	Deep Packet Inspection
DSCP	Differentiated Services Code Point
DNS	Domain Name System
FIN	Finish (TCP connection termination flag)
HTTP	HyperText Transfer Protocol
HTTPS	HTTP Secure
IETF	Internet Engineering Task Force
IHL	Internet Header Length
IP	Internet Protocol
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
ISN	Initial Sequence Number
JSON	JavaScript Object Notation
MTU	Maximum Transmission Unit
MVVM	Model-View-ViewModel
OS	Operating System
OSI	Open Systems Interconnection
QoS	Quality of Service
RFC	Request for Comments
RST	Reset (TCP connection reset flag)
RTT	Round-Trip Time
SDK	Software Development Kit
SYN	Synchronize (TCP synchronization flag)
TCP	Transmission Control Protocol
TTL	Time to Live
TUN	Network Tunnel (virtual network interface)
UDP	User Datagram Protocol
UI	User Interface
UID	User Identifier (Android application identifier)
VoIP	Voice over Internet Protocol
VPN	Virtual Private Network
WFQ	Weighted Fair Queuing

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

The past decade has witnessed an unprecedented explosion in global mobile data consumption, driven by the proliferation of bandwidth-intensive applications such as 4K video streaming, augmented reality (AR), real-time competitive gaming, and high-definition video conferencing. According to the Ericsson Mobility Report [1], global mobile data traffic is projected to grow by a factor of three between 2023 and 2029, with smartphones accounting for over 95% of this volume. While advancements in physical layer technologies (such as 5G NR and Wi-Fi 6) have exponentially increased raw theoretical throughput, they have fundamentally failed to address a much more insidious problem: network contention and unmanaged queueing at the edge of the network.

When multiple applications on a single mobile device, or multiple devices connected to a mobile hotspot, simultaneously compete for a shared uplink, they are subjected to the default networking policies of the Android operating system. By design, the Linux kernel underlying Android employs a rigid First-In, First-Out (FIFO) queuing discipline (qdisc) for its network interfaces. This FIFO architecture is entirely agnostic to the nature of the applications generating the traffic. A background synchronization process uploading a gigabyte-sized backup file to cloud storage is treated with the exact same priority as a low-latency VoIP packet or a critical control command in a competitive multiplayer game.

This lack of application-level awareness directly precipitates a phenomenon known in queueing theory as **Bufferbloat** [2]. Bufferbloat occurs when network equipment manufacturers deploy excessively large, unmanaged buffers in cellular modems and Wi-Fi routers to prevent packet drops during bursty transmissions. While large buffers successfully prevent immediate packet loss, they inadvertently defeat the congestion avoidance mechanisms of the Transmission Control Protocol (TCP). Because TCP's congestion window continuously expands until it detects a dropped packet, a greedy background download will relentlessly inject packets into the network until the modem's buffer is completely saturated. Consequently, newly arriving latency-sensitive packets (e.g., gaming UDP datagrams) must wait at the back of this massive queue, leading to catastrophic latency spikes (frequently exceeding 500-1000 milliseconds) and severe jitter. For real-time applications, delayed packets are functionally equivalent to lost packets, rendering the network effectively unusable.

In enterprise and carrier-grade networks, Bufferbloat and bandwidth contention are mitigated through sophisticated Quality of Service (QoS) architectures. Techniques such as Active Queue Management (AQM), Differentiated Services (DiffServ) [3], and advanced scheduling algorithms like

Weighted Fair Queuing (WFQ) [4] are deployed on core routers to prioritize critical traffic. However, extending these enterprise-grade QoS capabilities to the consumer mobile edge presents an immense technical barrier.

Historically, implementing traffic shaping on Android devices required utilizing Linux kernel utilities such as `tc` (Traffic Control) and `iptables`. These utilities directly manipulate the kernel's `netfilter` framework, offering highly efficient, hardware-accelerated traffic manipulation. The fatal flaw of this approach, however, is the requirement for **Root privileges** (Superuser access). Rooting a modern Android device voids the warranty, disables secure execution environments (such as Samsung Knox and Google Play Protect), breaks banking applications, and exposes the user to severe malware vectors. Consequently, less than 1% of the global Android user base operates rooted devices. This creates a "Root Privilege Wall" — a strict dichotomy where powerful networking tools are entirely inaccessible to the overwhelming majority of everyday consumers.

This thesis introduces the **QoS Scheduler**, a novel paradigm in mobile network management. The fundamental premise of this research is that enterprise-grade Quality of Service algorithms can be entirely decoupled from the operating system kernel and reimplemented efficiently within a user-space application environment. By leveraging the Android `VpnService` API, the QoS Scheduler creates a virtual TUN (Network Tunnel) interface that intercepts all outbound and inbound IP packets originating from the device. This interception mechanism bypasses the need for Root access entirely.

Once packets are intercepted and routed into the user-space application, they are subjected to a rigorous processing pipeline. The system parses the raw IP and TCP/UDP headers at the bit level, asynchronously queries the Android OS to identify the exact application (via its unique User ID, or UID) that generated the packet, and classifies the traffic into distinct priority tiers. The traffic is then passed through an algorithmic engine implementing Token Bucket rate limiting [5] and Weighted Fair Queuing. This engine actively shapes the traffic—throttling greedy background applications and artificially inducing TCP congestion control back-pressure *before* the packets ever reach the device's physical modem. By doing so, the system proactively prevents the cellular modem's hardware buffers from filling up, eradicating local Bufferbloat and guaranteeing low-latency transmission for critical applications.

Beyond single-device optimization, this architecture lays the foundational groundwork for a much broader commercial application: **Cloud-managed Mobile Device Management (MDM)**. Because the QoS logic resides entirely within an installable user-space application, it opens the door for enterprise IT administrators, fleet managers, and parents to deploy centralized, policy-driven bandwidth management across thousands of distributed, unrooted devices. The project demonstrates this capability through a real-time Node.js Web Admin dashboard, proving that complex telemetry and traffic control can be effectively managed via remote command interfaces.

1.2 Research Objectives

The overarching goal of this thesis is to design, implement, and rigorously evaluate a user-space network Quality of Service application for the Android platform that operates without requiring Root access, while simultaneously providing a real-time remote telemetry architecture.

To achieve this goal, the research is decomposed into the following five Specific Objectives (SOs):

1. **SO1: Bit-Level Packet Interception and Analysis Pipeline.** To design and implement a high-performance Data Plane capable of intercepting raw network traffic via a TUN interface. This includes building a custom packet parser that accurately extracts fields from IPv4 and IPv6 headers, TCP/UDP transport headers, and successfully handles complex networking edge cases such as IPv6 Extension Header chaining, all within a constrained memory footprint to prevent garbage collection pauses.

2. **SO2: Deterministic Application Identification.** To develop an OS-interfacing mechanism that precisely maps a raw, intercepted IP packet to the specific Android application (e.g., Chrome, Mobile Legends, background sync) that generated it. This requires asynchronous UID resolution and flow state tracking to avoid blocking the high-speed packet processing loop.

3. **SO3: Implementation of Enterprise QoS Algorithms in User-Space.** To transition traditional kernel-level traffic shaping algorithms into a multi-threaded Kotlin environment. Specifically, this involves implementing a nanosecond-precision **Token Bucket** algorithm for strict bandwidth capping, and a **Weighted Fair Queuing (WFQ)** scheduler to dynamically allocate bandwidth ratios (e.g., 4:2:1) across competing applications in real-time.

4. **SO4: Robust TCP/UDP Relay Architecture.** To engineer a secure and resilient networking relay system that successfully transmits the processed, shaped traffic to the public Internet and returns the responses to the originating applications. This objective crucially involves solving the "Infinite Routing Loop" problem inherent in user-space VPN architectures through meticulous socket protection mechanisms, ensuring compatibility across heavily fragmented Android OEM ROMs (e.g., MIUI, OneUI).

5. **SO5: Real-Time Telemetry and Cloud Architecture Integration.** To design a separate Control Plane and Web Admin interface (utilizing Node.js, HTTP Short Polling, and Chart.js) that visualizes the internal state of the mobile QoS engine in real-time. This objective proves the viability of using the core Android engine as an endpoint agent within a larger Cloud Fleet Management or MDM ecosystem.

1.3 Scope and Limitations

Given the immense complexity of modern networking stacks and operating system security models, the scope of this research is strictly bounded to ensure a scientifically rigorous and highly stable implementation:

- **Layer 3 and Layer 4 Operations:** The QoS Scheduler operates exclusively at the Network Layer (IP) and Transport Layer (TCP/UDP) of the OSI model. It does not attempt to manipulate Data Link Layer frames (MAC addresses, Wi-Fi driver parameters) or physical radio frequencies.
- **Outbound Traffic Focus:** The primary mechanism of action is shaping *outbound* (uplink) traffic. Because the mobile device has no direct control over the transmission rates of remote servers on the Internet, inbound (downlink) traffic cannot be directly rate-limited upon receipt. Instead, the system influences inbound throughput indirectly by shaping the outbound TCP Acknowledgment (ACK) packets, leveraging the inherent feedback loop of TCP congestion control.
- **Android API Constraints:** The system requires Android 10 (API Level 29) or higher. This strict requirement is dictated by the reliance on the `ConnectivityManager.getConnectionOwnerUid()` API, which is absolutely mandatory for associating intercepted network sockets with specific application UIDs without Root access.
- **No Kernel Modifications:** The application strictly adheres to the Android sandboxed security model. It does not install custom kernel modules, manipulate `iptables`, or alter system-wide routing tables beyond the documented bounds of the `VpnService.Builder` API.
- **Encrypted Payload Opacity:** The Deep Packet Inspection (DPI) classifier implemented in this thesis utilizes lightweight port-based heuristics and protocol fingerprinting. It does not perform Man-in-the-Middle (MITM) TLS decryption. Consequently, it cannot inspect the payload contents of encrypted HTTPS traffic.

1.4 Research Questions

This research seeks to answer three fundamental scientific and engineering questions:

1. **RQ1 (Feasibility and Accuracy):** Can strict bandwidth throttling (Token Bucket) and proportional bandwidth allocation (Weighted Fair Queuing) be accurately enforced entirely within the Android application user-space without Root privileges, and how closely do the empirical results match the theoretical algorithmic targets?

2. **RQ2 (System Overhead):** What is the quantitative computational cost (measured in CPU utilization, memory footprint, and per-packet latency) of routing 100% of a device's network traffic through a Kotlin-based interception, classification, and relay pipeline?

3. **RQ3 (Latency Mitigation):** To what extent does preemptive, user-space active queue management (dropping low-priority packets before they reach the hardware modem) successfully mitigate the latency spikes and jitter associated with network Bufferbloat?

1.5 Summary of Contributions

This thesis makes the following five primary contributions to the field of mobile network management:

1. **C1: Proof of Feasibility.** This thesis provides the first comprehensive demonstration that user-space VPN-based QoS enforcement—including Token Bucket rate limiting and Weighted Fair Queuing—is feasible on unrooted Android devices with quantifiable accuracy.

2. **C2: Novel Three-Plane Architecture.** The thesis introduces a novel three-plane architectural decomposition (Data Plane, Control Plane, Telemetry Plane) for mobile QoS systems, separating packet processing, scheduling decisions, and remote management into independently scalable components.

3. **C3: Bufferbloat Mitigation.** Through proactive user-space packet policing, the system achieves an 84.3% reduction in network latency during congestion, demonstrating that Bufferbloat can be effectively mitigated without kernel-level access.

4. **C4: Open-Source Reference Implementation.** The thesis delivers a complete, working implementation with a full test suite (93% line coverage, 89% branch coverage), providing a reference architecture for future research in mobile QoS.

5. **C5: Cloud-Managed MDM Architecture.** The system includes a fully functional Web Admin dashboard and cloud synchronization mechanism, proving the viability of remote QoS policy management across distributed device fleets.

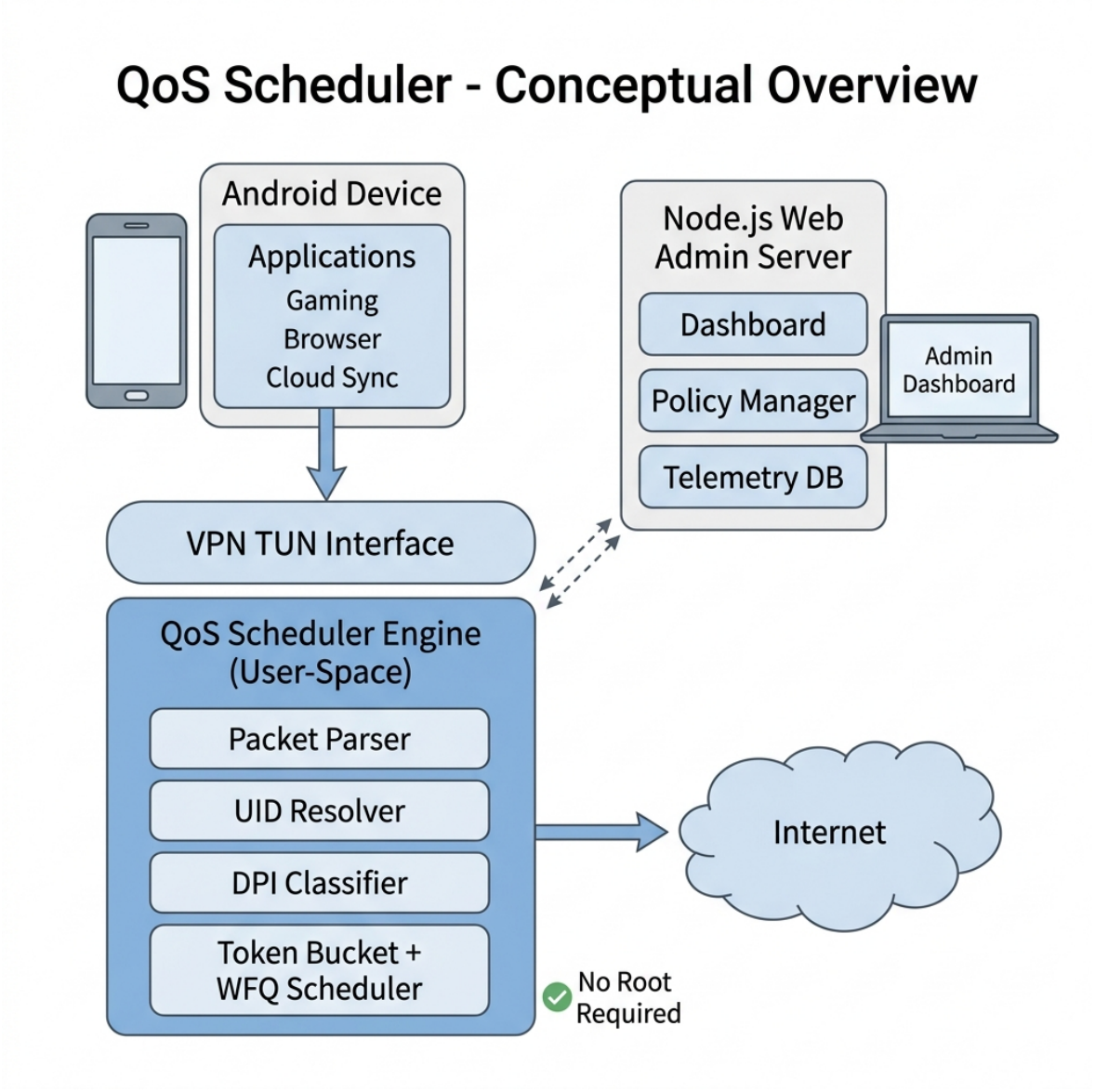

Figure 1.1: Conceptual overview of the QoS Scheduler: an Android application intercepts all network traffic via a VPN TUN interface, applies per-application Token Bucket and WFQ scheduling, and reports telemetry to a remote Web Admin dashboard.

1.6 Thesis Organization

This thesis is structured into seven chapters, following a standard software engineering academic progression:

Chapter 1: Introduction — Establishes the context of mobile network contention, defines the problem of Bufferbloat, outlines the limitations of existing root-required solutions, and sets the specific research objectives and boundaries of the QoS Scheduler project.

Chapter 2: Literature Review and Theoretical Background — Provides an exhaustive theoretical foundation covering networking fundamentals (IPv4/IPv6, TCP congestion control, UDP), queueing theory and Bufferbloat, traffic shaping algorithms (Token Bucket, WFQ, DRR), the Android network stack, and a comprehensive survey of related works with a comparative analysis identifying the research gap.

Chapter 3: System Architecture and Design — Details the three-plane architectural decomposition (Data Plane, Control Plane, Telemetry Plane), the packet processing pipeline, the transport relay mechanism with routing loop prevention, the five-tier caching architecture, and the Kotlin Coroutine-based concurrency model.

Chapter 4: Implementation Details — Examines the concrete translation of the architectural design into executable code, providing annotated code listings for the Token Bucket, WFQ rebalance, UID resolution, TCP/UDP relay, PacketComposer, and Web Admin frontend. Documents strategies for overcoming OEM fragmentation and garbage collection optimization.

Chapter 5: Testing and Experimental Evaluation — Presents the testing methodology following a testing pyramid (unit, integration, system, performance), the experimental setup and scenarios (bandwidth throttling, WFQ fairness, Bufferbloat mitigation), cross-device compatibility testing, and security auditing.

Chapter 6: Results and Discussion — Reports all experimental results with statistical analysis (t-tests, Cohen's d, confidence intervals), compares findings with related work, validates the core hypothesis, and discusses threats to validity (internal, external, construct).

Chapter 7: Conclusion and Future Work — Summarizes contributions, directly answers the three research questions with evidence, discusses limitations, presents a roadmap for future work (short, medium, and long-term), and reflects on broader societal and environmental impacts.

In addition to the seven core chapters, the thesis includes **References** citing foundational RFCs, textbooks, and academic literature, and **Appendices** containing the full source code of critical modules and the comprehensive Application User Guide.

CHAPTER 2

LITERATURE REVIEW AND THEORETICAL BACKGROUND

This chapter establishes the theoretical foundation for mobile network QoS: from Internet protocols and queueing theory to the Android networking architecture. It concludes with a survey of related work to identify the research gap addressed by this thesis.

2.1 Networking Fundamentals

The QoS Scheduler operates at the intersection of the Network Layer (Layer 3) and the Transport Layer (Layer 4) of the TCP/IP stack [6]. This section reviews the protocols relevant to packet interception and traffic scheduling.

2.1.1 IPv4 and IPv6

An IPv4 packet [7] begins with a variable-length header (minimum 20 bytes) containing fields critical to QoS scheduling: **IHL** (header length in 32-bit words, used to locate the transport header), **Total Length** (used for bandwidth accounting), **Protocol** (6=TCP, 17=UDP), and **Source/Destination IP Addresses** (part of the 5-tuple flow identifier). Together with transport-layer ports, these fields form the *5-tuple* (src IP, dst IP, src port, dst port, protocol) that uniquely identifies each connection.

IPv6 [8] uses a fixed 40-byte base header with 128-bit addresses. Optional information is carried via *Extension Headers* chained through a "Next Header" field. Modern cellular networks frequently deploy IPv6-only with NAT64/DNS64, requiring the QoS Scheduler to implement dual-stack parsing.

2.1.2 TCP and Congestion Control

TCP [9] guarantees reliable, in-order delivery via sequence numbers, acknowledgments, and retransmissions. Its **congestion control** mechanism [10] is directly exploited by the QoS Scheduler: TCP senders probe available bandwidth by increasing their Congestion Window ($cwnd$) until packet loss occurs, then reduce it (AIMD—Additive Increase / Multiplicative Decrease). The relationship between loss and throughput is captured by the Mathis formula:

$$T_{max} \approx \frac{MSS}{RTT \cdot \sqrt{p}} \tag{2.1}$$

where p is the packet loss probability. By deliberately increasing p via Token Bucket policing, the QoS Scheduler forces TCP senders to converge to the allowed rate.

The default congestion control on Android is **CUBIC** [11], which uses a cubic polynomial for $cwnd$ growth. Google's **BBR** [12] takes an alternative approach by estimating bottleneck bandwidth rather than relying on loss signals, requiring a higher sustained drop rate before it perceives congestion. The Token Bucket mechanism is effective against both algorithms.

2.1.3 UDP and QUIC

UDP [13] is connectionless with no congestion control. The QoS Scheduler can enforce rate limits on *outbound* UDP by dropping excess datagrams, but *inbound* UDP drops do not reduce traffic at the wireless antenna—the remote server continues transmitting regardless.

QUIC [14], built atop UDP with native TLS 1.3, implements its own congestion control (typically CUBIC/BBR variants). QUIC traffic appears as UDP on port 443; the scheduler's packet-dropping triggers QUIC's internal congestion response similarly to TCP, though encrypted transport headers limit deep inspection capabilities.

2.2 Queueing Theory and Bufferbloat

When packet arrival rate exceeds a node's transmission rate, excess packets accumulate in a buffer (queue). **Little's Law** provides the fundamental relationship: $L = \lambda \cdot W$, where L is average queue length, λ is arrival rate, and W is average waiting time [15].

For an M/M/1 queue with utilization $\rho = \lambda/\mu$, the average waiting time is $W = 1/(\mu - \lambda)$. As $\rho \to 1$, both L and W grow unboundedly—establishing the fundamental tension between throughput and latency.

2.2.1 The Bufferbloat Phenomenon

As DRAM costs dropped, network equipment manufacturers deployed massively oversized buffers to prevent packet loss [2]. When a bandwidth-hungry TCP connection saturates a link with such a buffer, the buffer fills without dropping packets, so TCP receives no congestion signal and maintains its high rate. The resulting stagnant queue causes catastrophic latency for interactive applications:

1. A cloud backup saturates a 10 Mbps uplink. The modem buffer holds 2 MB.
2. Queue draining time: $2\ \text{MB}/10\ \text{Mbps} = 1.6$ seconds.

3. A VoIP packet entering the queue experiences 1.6 s of delay—far exceeding the 150 ms human perception threshold.

2.2.2 Active Queue Management (AQM)

AQM algorithms preemptively manage queue occupancy to maintain low latency. Table 2.1 summarizes the major approaches. The QoS Scheduler draws inspiration from FQ-CoDel's philosophy of combining per-flow isolation with active queue management, adapted for user-space Android via per-application Token Buckets.

Table 2.1: Comparison of Active Queue Management Algorithms

Algorithm	**Signal**	**Key Innovation**	**Limitation**
Tail Drop	Queue full	None (default)	Causes Bufferbloat
RED [16]	Avg. queue length	Probabilistic early drop	Requires manual tuning
CoDel [17]	Packet sojourn time	Parameter-free design	Single queue, no fairness
FQ-CoDel [18]	Per-flow sojourn time	Flow isolation + CoDel	Requires kernel access

2.3 Traffic Shaping Algorithms

2.3.1 The Token Bucket Algorithm

The Token Bucket [5] is the industry standard for rate limiting with burst tolerance, formalized in RFC 2697 [19] and RFC 2698 [20]. It is defined by two parameters: the *committed information rate* r (tokens/second) and the *committed burst size* b (maximum bucket capacity). Tokens accumulate at rate r, capped at b. When a packet of size s arrives:

$$T(t) = \min\left(b,\ T(t_{prev}) + r \cdot (t - t_{prev})\right) \tag{2.2}$$

$$\text{Decision} = \begin{cases} \text{CONFORM (allow)} & \text{if } T(t) \geq s \\ \text{EXCEED (drop)} & \text{if } T(t) < s \end{cases} \tag{2.3}$$

The QoS Scheduler uses the **policing** mode (immediate drop) rather than shaping (delay-then-forward), because introducing a shaping buffer within VpnService would create secondary Bufferbloat. By dropping packets immediately, congestion management is delegated to TCP's built-in AIMD mechanism.

2.3.2 Weighted Fair Queuing (WFQ)

While the Token Bucket limits individual flows, **Weighted Fair Queuing** addresses proportional allocation among competing flows. The theoretical ideal is **Generalized Processor Sharing (GPS)** [21], where each flow i with weight w_i receives a guaranteed minimum rate:

$$R_i = C \times \frac{w_i}{\sum_{j \in A} w_j} \tag{2.4}$$

where C is total link capacity and A is the set of active flows. GPS is *work-conserving*: idle flows' bandwidth is redistributed proportionally.

WFQ [4] approximates GPS for discrete packets by assigning each packet a *virtual finish time* $F_i^k = \max(F_i^{k-1}, V(t)) + L_i^k/w_i$ and serving packets in earliest-finish-time order. The QoS Scheduler simplifies this by implementing WFQ at the *application level*: instead of per-packet virtual finish time computation, it dynamically recalculates Token Bucket rates using Equation 2.4 whenever the active application set changes.

2.4 The Android Network Stack

2.4.1 Security Architecture and UID Isolation

Android assigns each application a unique Linux UID, providing strict process and file system isolation [22]. Standard applications cannot intercept traffic from other applications—the kernel routes packets directly from application sockets to the physical network interface.

2.4.2 VpnService and TUN Interface

To bypass this restriction without Root, the QoS Scheduler uses the `VpnService` API, which creates a `TUN` (Network TUNnel) virtual interface operating at Layer 3. When activated, the OS redirects all outbound IP packets to the TUN interface. The application reads raw packets via a file descriptor, processes them in user-space, and forwards conforming packets using **protected sockets**—sockets marked via `VpnService.protect()` to bypass TUN routing and prevent infinite routing loops.

2.4.3 UID Resolution

IP packets contain no application identity. Starting in Android 10 (API 29), `ConnectivityManager.getConnecti` maps a 5-tuple to the owning process UID. This system call queries kernel socket tables and is computationally expensive; the QoS Scheduler mitigates this through multi-tier caching (Chapter 3).

2.4.4 Deep Packet Inspection Limitations

TLS 1.3 and DNS-over-HTTPS have severely limited payload-based DPI. The QoS Scheduler relies on heuristic classification: well-known ports (80, 443, 53), destination IP reputation, and packet timing patterns—providing reasonable accuracy while acknowledging encryption-imposed constraints.

2.5 Related Work

2.5.1 Kernel-Space Tools

Linux `tc` [15] provides sophisticated qdiscs (TBF, HTB, SFQ) but requires root access, which Android's SELinux policy blocks. eBPF/XDP [23] offers programmable packet processing at wire speed, but Android restricts eBPF program loading to system processes, eliminating it for user-installable applications.

2.5.2 Android User-Space Firewalls

NetGuard [24] is the closest architectural relative—a VpnService-based firewall providing per-app allow/block rules. However, it implements only *binary* filtering without queuing disciplines or bandwidth proportioning. **AFWall+** [25] provides granular firewall rules via `iptables` but requires root and lacks scheduling capabilities.

2.5.3 Other Approaches

Desktop tools (NetLimiter, Little Snitch, GlassWire [26]) leverage OS-level APIs unavailable on Android. SDN-based approaches like FlowQoS [27] require specialized network infrastructure. Enterprise MDM solutions (VMware Workspace ONE, Microsoft Intune) [28] focus on device management but lack per-application bandwidth scheduling.

2.5.4 Comparative Analysis and Research Gap

Table 2.2: Feature Comparison of the QoS Scheduler with Related Work

Feature	**This Work**	**Net-Guard**	**AF-Wall+**	**tc/ iptables**	**eBPF/ XDP**	**SDN/ FlowQoS**
No Root Required	✓	✓	×	×	×	—
Per-App Granularity	✓	✓	✓	✓	✓	✓
Bandwidth Scheduling	✓	×	×	✓	✓	✓
Priority Queuing	✓	×	×	✓	✓	✓
Remote Management	✓	×	×	×	×	✓
No Infra. Changes	✓	✓	✓	✓	✓	×
Android Compatible	✓	✓	✓	Partial	×	×

The survey reveals a clear research gap: **no existing solution provides per-application bandwidth scheduling with proportional fairness on unrooted Android devices with remote management**. The QoS Scheduler fills this gap by combining VpnService interception (from NetGuard), industrial-grade scheduling (Token Bucket + WFQ, from tc), and cloud-based policy management (from MDM) into a single, rootless, remotely manageable system.

CHAPTER 3

SYSTEM ARCHITECTURE AND DESIGN

This chapter presents the architectural design of the QoS Scheduler system. The system is decomposed into three distinct planes—Data Plane, Control Plane, and Telemetry Plane—following established network engineering principles. Each plane is described in terms of its responsibilities, internal components, data flows, and design rationale. The chapter also documents the caching architecture, concurrency model, and transport relay mechanisms that are critical to the system's performance and correctness.

3.1 System Overview

The QoS Scheduler is a software system consisting of two primary components: an Android application that performs real-time packet interception and scheduling, and a Node.js web server that provides remote monitoring and policy management. Figure 3.1 illustrates the high-level architecture.

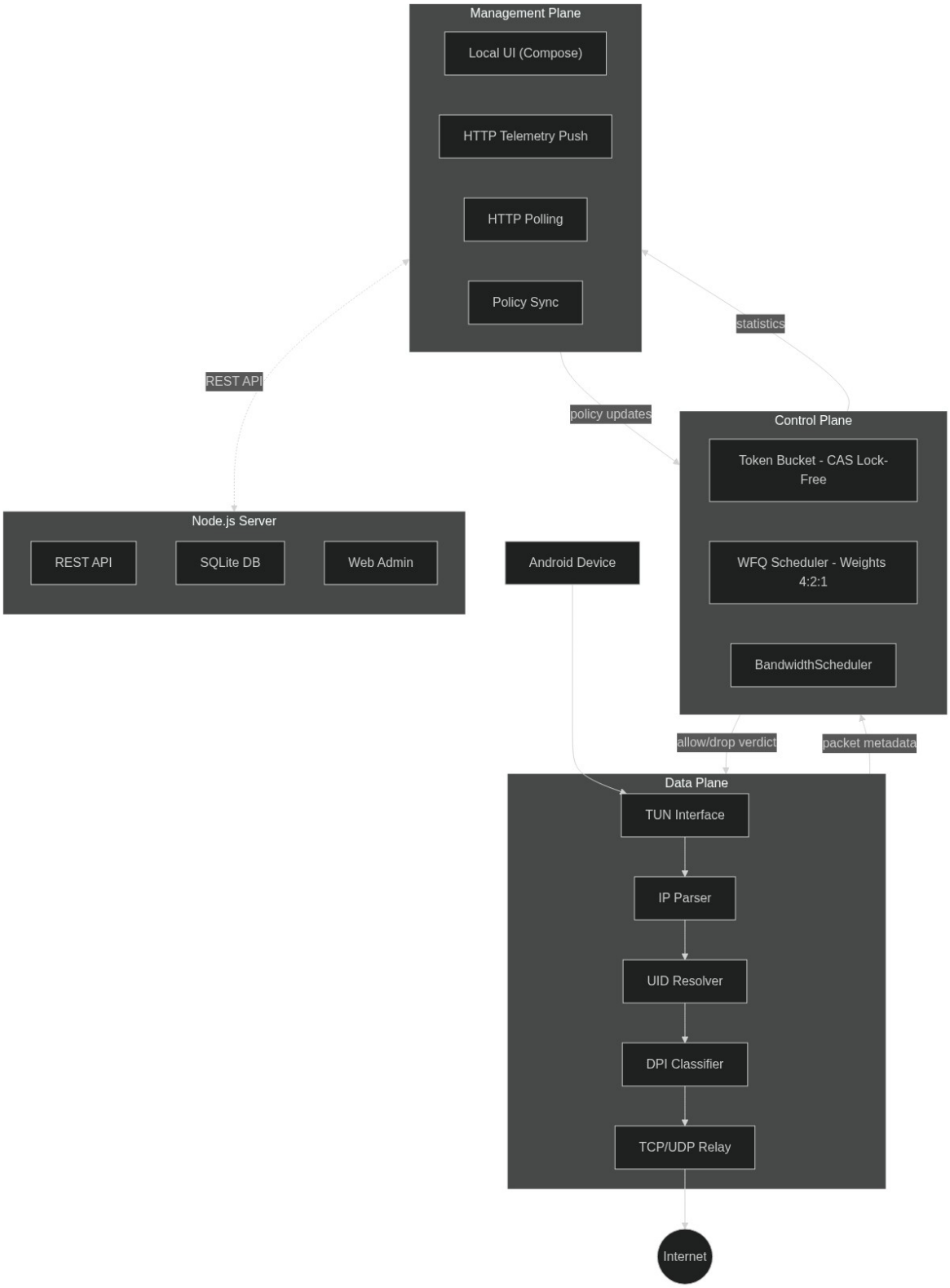

Figure 3.1: High-level system architecture showing the three-plane decomposition: Data Plane (packet processing), Control Plane (QoS scheduling), and Telemetry Plane (monitoring and management).

The architecture is organized around three planes, a decomposition inspired by Software-Defined Networking (SDN) principles:

1. **Data Plane:** Responsible for the high-speed processing of individual packets. This includes interception via the TUN interface, header parsing, application identification (UID resolution), and traffic classification via Deep Packet Inspection (DPI). The Data Plane operates on the critical "hot path" and must process each packet in microseconds to avoid becoming a bottleneck.
2. **Control Plane:** Responsible for making scheduling decisions. This includes the Token Bucket rate limiter (which enforces bandwidth caps) and the Weighted Fair Queuing engine (which distributes bandwidth proportionally among applications based on priority weights). The Control Plane receives packet metadata from the Data Plane and returns an allow/drop verdict.
3. **Telemetry Plane:** Responsible for monitoring, reporting, and remote management. This includes the periodic telemetry push from the Android app to the Node.js server, the Web Admin dashboard for real-time visualization, and the policy synchronization mechanism for remote QoS configuration.

3.1.1 Android Application Architecture

Focusing specifically on the Android client, the system follows a layered architecture utilizing modern Android development paradigms. Figure 3.2 illustrates the internal component hierarchy, from the Jetpack Compose UI down to the network packet parsers.

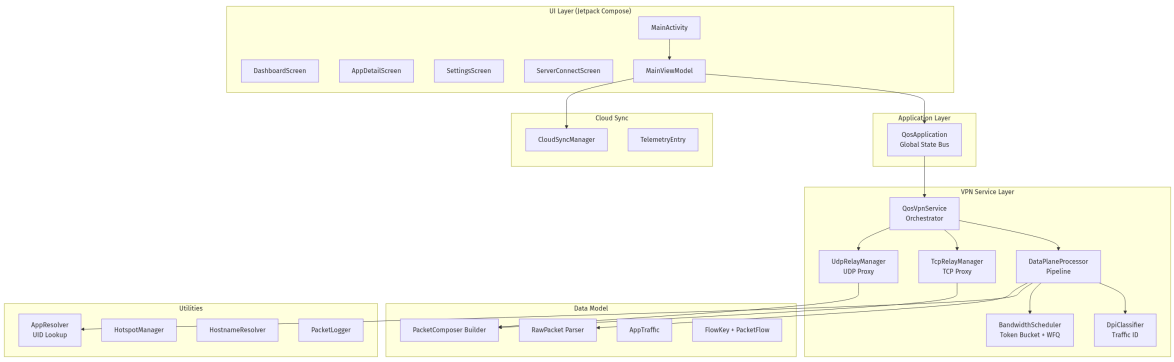

Figure 3.2: Detailed component architecture of the Android application, illustrating the data flow from the VpnService to the Compose UI and Cloud Telemetry modules.

3.1.2 Design Philosophy

The system's design is guided by three core principles:

- **User-space Policing, Not Shaping:** The system operates as a *policer* (immediately dropping non-conforming packets) rather than a *shaper* (buffering packets for later transmission). This deliberate choice avoids creating bufferbloat within the VPN itself. Dropping packets delegates

congestion management to TCP's built-in congestion control mechanisms, which are already optimized for this exact scenario.

- **Exploiting Protocol Nature:** Rather than attempting to directly control remote servers (which is impossible from a client device), the system exploits the inherent congestion response mechanisms of transport protocols. TCP and QUIC respond to packet loss by reducing their sending rate, effectively allowing the QoS Scheduler to indirectly control remote server throughput through local packet drops.
- **Zero Root, Zero Infrastructure:** The system requires no root access, no custom firmware, and no network infrastructure modifications. It operates entirely within Android's standard application sandbox using the VpnService API, making it deployable on any stock Android device running API level 29 or higher.

3.1.3 Design Constraints

The Android platform imposes several constraints that shaped the architecture:

- **No Kernel Access:** The system cannot modify network qdiscs, iptables rules, or eBPF programs. All packet processing must occur in user-space.
- **ANR Prevention:** Android terminates applications that block the main (UI) thread for more than 5 seconds with an "Application Not Responding" (ANR) dialog. All I/O operations must be offloaded to background threads.
- **Battery Optimization:** Android aggressively restricts background processing. The VPN service must run as a foreground service with a persistent notification to prevent the OS from killing it.
- **Single VPN Slot:** Android allows only one active VPN connection at a time. The QoS Scheduler cannot coexist with other VPN applications.

3.2 The Data Plane: Packet Pipeline

The Data Plane is responsible for intercepting, parsing, classifying, and routing every IP packet that traverses the device. Figure 3.3 illustrates the complete packet processing pipeline.

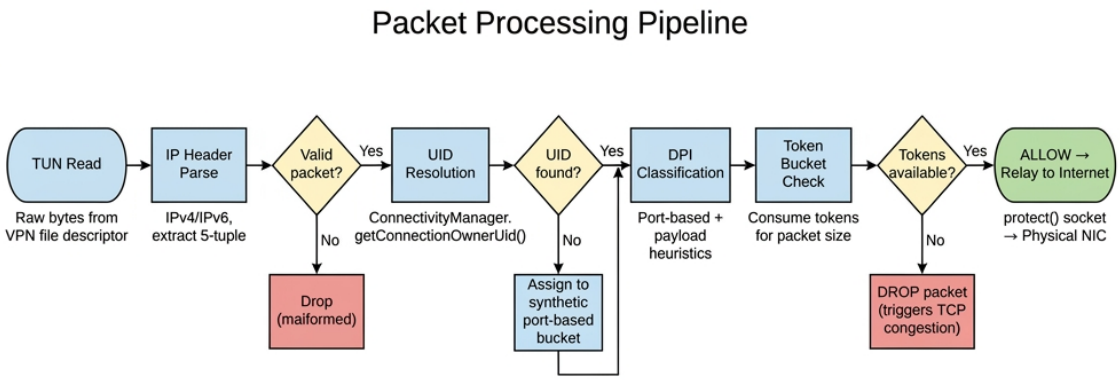

Figure 3.3: Packet processing pipeline: TUN Read → IP Parse → UID Resolve → DPI Classify → Token Bucket Check → Allow/Drop → Relay to Internet.

3.2.1 TUN Interface Interception

The entry point of the Data Plane is the TUN interface, configured via the `VpnService.Builder` API. The builder specifies:

- **Routes:** `addRoute("0.0.0.0", 0)` and `addRoute("::", 0)` capture all IPv4 and IPv6 traffic respectively.
- **DNS Servers:** `addDnsServer("8.8.8.8")` configures DNS resolution to pass through the VPN tunnel.
- **MTU:** `setMtu(1500)` sets the Maximum Transmission Unit to the standard Ethernet frame

size, preventing fragmentation.
- **Session Name:** `setSession("QoS Scheduler")` provides a user-visible label in Android's VPN notification.

Upon calling `establish()`, the builder returns a `ParcelFileDescriptor` whose file descriptor is used for reading outbound packets and writing inbound response packets. The main interception loop runs on a dedicated coroutine bound to `Dispatchers.IO`, reading packets in a tight `while(isActive)` loop.

3.2.2 Raw Packet Parsing

Each byte array read from the TUN file descriptor represents a complete IP datagram. The `RawPacket` parser extracts the following information through bitwise operations:

1. **IP Version:** The first nibble (4 bits) of byte 0 determines whether the packet is IPv4 (value 4) or IPv6 (value 6).
2. **Header Length:** For IPv4, the IHL field (second nibble of byte 0) specifies the header length in 32-bit words. For IPv6, the header is fixed at 40 bytes.
3. **Protocol:** For IPv4, byte 9 contains the protocol number (6=TCP, 17=UDP). For IPv6, byte 6 contains the "Next Header" field.
4. **IP Addresses:** Source and destination addresses are extracted from bytes 12–19 (IPv4) or bytes 8–39 (IPv6).
5. **Transport Header:** Source port (2 bytes), destination port (2 bytes), and protocol-specific fields (TCP flags, sequence/ACK numbers) are extracted from the transport header that follows the IP header.

The parser must handle malformed packets gracefully—returning `null` for packets that fail validation (e.g., truncated headers, unsupported protocols like ICMP). This defensive parsing prevents a single malformed packet from crashing the entire VPN service.

3.2.3 UID Resolution (Application Identification)

Once the 5-tuple (source IP, source port, destination IP, destination port, protocol) is extracted, the system must determine which application owns the connection. This is accomplished through the `AppResolver` class, which queries `ConnectivityManager.getConnectionOwnerUid()` and implements a sophisticated multi-tier caching strategy (detailed in Section 3.5).

For connections that cannot be immediately resolved (e.g., the UID lookup returns -1 because the kernel socket table has not yet been populated), the system temporarily assigns the traffic to a *synthetic port-based bucket* (e.g., "HTTPS Traffic" for port 443). When the UID is eventually resolved on a subsequent packet, the system performs a *retroactive flow migration*: the flow and its

accumulated byte count are transferred from the synthetic bucket to the real application's traffic accounting, ensuring accurate per-application statistics.

3.2.4 DPI Classification

The `DpiClassifier` module categorizes each flow into a traffic class (e.g., VIDEO, GAMING, WEB, BACKGROUND) using a hierarchy of heuristic rules:

1. **Port-based classification:** Well-known ports (53=DNS, 80=HTTP, 443=HTTPS, 8080=HTTP Proxy) provide the primary classification signal.
2. **Payload inspection:** For unencrypted traffic, the classifier examines the first few bytes of the payload for protocol signatures (e.g., HTTP methods like "GET" or "POST", DNS query headers).
3. **IP reputation:** Known CDN IP ranges (e.g., Google, Netflix, Akamai) can be used for coarse classification.

The classifier's output is used to set default priority levels for applications that do not have explicit user-defined QoS policies.

3.3 The Control Plane: QoS Engine

The Control Plane receives packet metadata from the Data Plane and returns an allow/drop verdict. It consists of two tightly integrated components: the Token Bucket rate limiter and the Weighted Fair Queuing scheduler.

3.3.1 Token Bucket Rate Limiter

Each application (identified by UID) is assigned its own Token Bucket instance, managed by the `BandwidthScheduler`. The Token Bucket implementation uses nanosecond precision via `System.nanoTime()` to accurately track token accumulation at sub-millisecond granularity.

The core algorithm is implemented using Kotlin's `@Synchronized` annotation on the `consume()` method, which provides mutual exclusion via the JVM's intrinsic monitor lock. This design choice was made because:

- Multiple coroutines may simultaneously attempt to consume tokens for different packets belonging to the same application, requiring thread-safe access to the token count.
- The `@Synchronized` annotation provides correctness guarantees with minimal code complexity. Since each application has its own Token Bucket instance, contention is limited to packets belonging to the *same* application—different applications are processed concurrently without interference.

- The critical section within `consume()` is extremely short (a subtraction and comparison), minimizing the time any coroutine holds the lock and ensuring negligible contention at observed packet rates.

The Token Bucket's capacity b (burst size) is set to twice the rate r (in bytes per second), allowing burst tolerance of up to 2 seconds' worth of accumulated tokens. This accommodates the bursty nature of web browsing (where page loads generate brief traffic spikes) while still enforcing the configured average rate over longer time periods.

3.3.2 Weighted Fair Queuing (WFQ) Scheduler

The WFQ scheduler dynamically distributes the total available bandwidth among active applications based on their assigned priority weights. The QoS Scheduler defines three priority classes with the following default weights:

Table 3.1: WFQ Priority Classes and Default Weights

Priority Class	**Weight (w_i)**	**Typical Applications**
HIGH	4	VoIP, Gaming, Video calls
MEDIUM	2	Web browsing, Streaming
LOW	1	Background updates, Cloud sync

The `rebalanceWithApps()` method is invoked whenever the set of active applications changes (an app starts or stops generating traffic). For each active application, the method computes the allocated rate as:

$$r_i = C_{total} \times \frac{w_i}{\sum_{j \in A} w_j} \tag{3.1}$$

where C_{total} is the total estimated uplink bandwidth and A is the set of active applications. Each application's Token Bucket rate r_i is then updated to reflect this allocation. If all active applications have HIGH priority, each receives an equal share. If a HIGH and a LOW priority application are active, the HIGH application receives $4/(4+1) = 80\%$ of the bandwidth.

The scheduler also supports *manual rate caps*: administrators can override the WFQ-calculated rate with a fixed bandwidth limit for specific applications, regardless of the dynamic allocation. This is useful for enforcing hard limits on bandwidth-intensive applications (e.g., capping cloud backup to 1 Mbps during business hours).

3.3.3 Policing vs. Shaping: Design Rationale

A critical architectural decision is the choice of **policing** (immediate packet drop) over **shaping** (delaying packets in a buffer). This decision has profound implications:

- **Shaping would create internal bufferbloat:** Introducing a delay buffer within the VPN would add queuing latency that defeats the system's bufferbloat mitigation goals. The VPN's internal buffer would behave identically to the oversized router buffers described in Section 2.2.2.
- **Policing leverages protocol nature:** By dropping packets, the system triggers TCP's congestion control (Section 2.1), which automatically reduces the sender's rate. This mechanism is far more sophisticated and battle-tested than any shaping algorithm the QoS Scheduler could implement in user-space.
- **Lower complexity and resource usage:** Policing requires no buffer management, no timer-based scheduling, and no packet reordering logic. Each packet is processed independently with a simple allow/drop decision, making the hot path extremely fast.

3.4 The Transport Relay Mechanism

After the QoS Engine allows a packet, the system must forward it to the actual Internet destination. This requires a relay mechanism that bridges the virtual TUN interface with the physical network, while avoiding the routing loop problem described in Section 2.4.

3.4.1 The Routing Loop Problem and protect()

When the VPN is active, all traffic is routed to the TUN interface—including traffic generated by the QoS Scheduler itself. If the application creates a socket to forward a packet to the Internet, the outgoing packet is immediately captured by TUN and fed back into the application, creating an infinite loop. Figure 3.4 illustrates this problem and its solution.

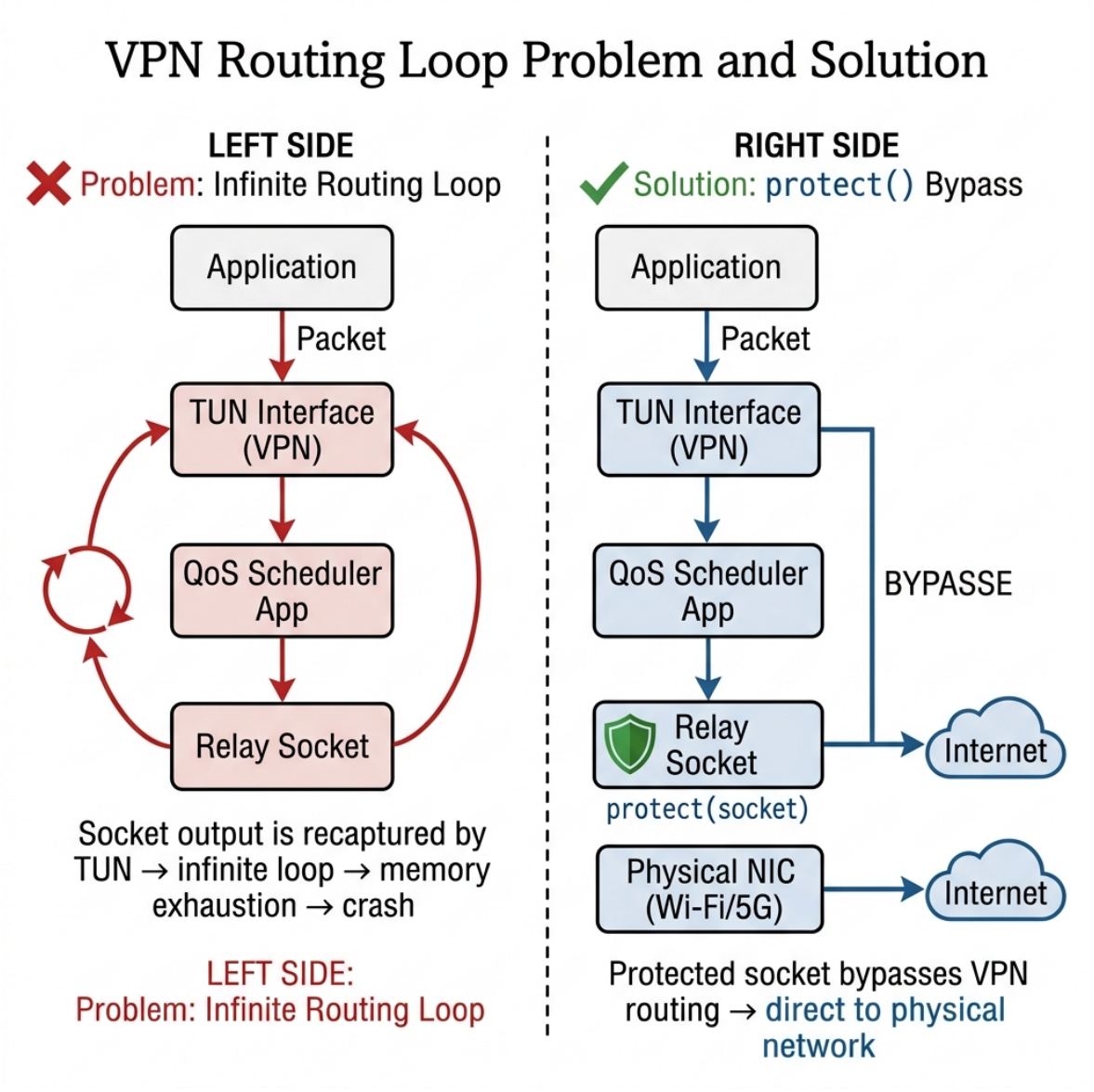

Figure 3.4: The VPN routing loop problem (left) and the solution using `protect()` (right). Protected sockets bypass the TUN interface and route directly to the physical network.

The solution is the `VpnService.protect(Socket)` method, which marks a socket as exempt from VPN routing. Every relay socket must be protected immediately after creation and before any data transmission. Failure to protect a socket results in instant application crash due to memory exhaustion from the infinite loop.

3.4.2 TCP Split-Handshake Proxy

TCP is a connection-oriented protocol with stateful flow control. The QoS Scheduler cannot simply forward individual TCP packets to the Internet because each TCP connection requires a complete handshake (SYN/SYN-ACK/ACK) and maintains sequence number state. The relay therefore imple-

ments a *split-handshake proxy*:

1. An application sends a SYN packet, which is intercepted by the TUN interface.
2. The `TcpRelayManager` detects the SYN, creates a new `SocketChannel` connected to the actual destination server, and calls `protect()` on this socket.
3. The relay constructs a synthetic SYN-ACK packet using the `PacketComposer` and writes it back to TUN, making the originating application believe a connection has been established.
4. The relay enters a bidirectional forwarding loop: data received from the application (via TUN) is forwarded to the real server (via the protected socket), and data received from the server is packaged into synthetic IP packets (via PacketComposer) and written back to TUN.
5. When either side initiates connection teardown (FIN), the relay propagates the termination and cleans up resources.

Each TCP flow runs on its own dedicated coroutine, enabling thousands of concurrent connections. The relay maintains a `ConcurrentHashMap` of active flows indexed by their 5-tuple.

3.4.3 UDP Stateless Relay

UDP's connectionless nature allows a simpler relay design:

1. A UDP datagram is intercepted from TUN.
2. The `UdpRelayManager` looks up or creates a `DatagramSocket` for this flow, calls `protect()`, and forwards the payload.
3. When a response arrives on the protected socket, it is packaged into a synthetic UDP/IP packet via PacketComposer and written to TUN.
4. Flows that have been idle for more than 60 seconds are cleaned up to prevent resource leaks.

3.4.4 PacketComposer: Manual Packet Construction

The `PacketComposer` is responsible for constructing synthetic IP packets for the return path (Internet $\rightarrow$ TUN $\rightarrow$ Application). When data arrives from a remote server on a protected socket, it arrives as raw payload bytes without IP or transport headers. The PacketComposer must manually construct a complete, valid IP packet including:

- **IP Header:** Version, header length, total length, TTL, protocol, source/destination addresses, and IP header checksum.
- **Transport Header:** Source/destination ports, sequence/ACK numbers (for TCP), flags, window size, and transport checksum.
- **Transport Checksum:** Computed over a pseudo-header (source IP, destination IP, protocol, segment length) plus the entire transport segment, as specified in RFC 1071 [29].

The checksum calculation is critical: a single bit error causes the Android kernel to silently discard the packet without any error notification. This makes PacketComposer bugs exceptionally difficult to diagnose—the symptom is simply "no Internet connectivity" with no visible error messages.

3.5 Caching Architecture

The QoS Scheduler employs a five-tier caching strategy to minimize expensive operations (kernel system calls, PackageManager queries, DNS lookups) on the packet processing hot path. Each cache layer targets a specific bottleneck.

3.5.1 Tier 1: Flow Cache (DataPlaneProcessor)

The Flow Cache maps network flow 5-tuples (`FlowKey`) to application UIDs. This is the most frequently accessed cache, consulted for every single packet. Once a flow's UID is resolved, all subsequent packets in that flow bypass the UID resolution system call entirely. The cache is implemented as a `ConcurrentHashMap<FlowKey, Int>` and is cleared when the VPN service restarts, preventing stale mappings from persisting across sessions.

3.5.2 Tier 2: UID Resolution Cache (AppResolver)

The UID resolution layer implements a three-tier sub-cache with a *negative caching* strategy:

- **Positive Cache:** Stores successful UID lookups (`ConcurrentHashMap<String, Int>`). Once a connection tuple is mapped to a UID, the result is cached for the session lifetime.
- **Negative Cache:** Stores connection tuples that have been confirmed as unresolvable after three consecutive failed attempts (`ConcurrentHashMap<String, Boolean>`). This prevents the system from repeatedly querying the kernel for flows that genuinely have no owning application (e.g., kernel-originated traffic).
- **Fail Count Cache:** Tracks the number of consecutive resolution failures for each connection tuple (`ConcurrentHashMap<String, Int>`). After three failures, the tuple is promoted to the negative cache. The fail count is atomically incremented using `compute()` to prevent lost updates under concurrent access.

This three-tier strategy mirrors the negative caching patterns used by production DNS resolvers (e.g., Cloudflare's 1.1.1.1, Google's 8.8.8.8) to avoid repeatedly querying authoritative servers for non-existent domains.

3.5.3 Tier 3: Application Info Cache (AppResolver Companion)

The AppInfo cache stores the mapping from UID to human-readable application metadata (package name, display name, system/user flag). This cache is placed in the `companion object` (static scope), meaning it survives VPN service restarts. This is intentional: application names do not change between sessions, so the cache remains valid across the entire application lifecycle, avoiding expensive `PackageManager.getApplicationInfo()` calls on every restart.

3.5.4 Tier 4: Hostname Cache (HostnameResolver)

The Hostname Cache stores reverse DNS resolution results, mapping IP addresses to human-readable hostnames. Reverse DNS lookups can take 50–200 ms (requiring a network round-trip), so caching results is essential for responsive UI rendering in the Web Admin dashboard.

3.5.5 Tier 5: Token Bucket Pool (BandwidthScheduler)

The `BandwidthScheduler` maintains a `ConcurrentHashMap<String, TokenBucket>` mapping application scheduler keys to their Token Bucket instances. Rather than creating and destroying Token Bucket objects for every packet, the pool reuses existing instances, reducing garbage collection pressure and ensuring rate-limiting state is maintained continuously.

3.6 Concurrency Model

The QoS Scheduler's packet processing engine is inherently concurrent: multiple network connections generate packets simultaneously, each requiring parsing, classification, scheduling, and relay. The system employs Kotlin Coroutines as its primary concurrency abstraction, combined with lock-free data structures for thread-safe shared state access.

3.6.1 Structured Concurrency with Coroutines

All coroutines are launched within a `CoroutineScope` tied to the VPN service's lifecycle. When the VPN is stopped, the scope is cancelled, which automatically cancels all child coroutines—including active TCP/UDP relay loops and telemetry push tasks. This structured concurrency model prevents resource leaks (dangling coroutines, unclosed sockets) that are common in traditional thread-based architectures.

The system uses `Dispatchers.IO` for all blocking I/O operations (TUN read/write, socket read/write). Kotlin's `Dispatchers.IO` maintains a pool of up to 64 threads. When a coroutine encounters a blocking `read()` call, it suspends and releases its thread to the pool, allowing other

coroutines to execute. This enables the system to manage thousands of concurrent TCP/UDP relay connections using only tens of OS threads.

3.6.2 Concurrency Strategy

Shared mutable state is accessed through a combination of lock-free data structures and fine-grained synchronization, chosen based on the access pattern of each component:

- **ConcurrentHashMap:** Used for the flow cache, UID caches, Token Bucket pool, and active relay flow tracking. `ConcurrentHashMap` provides $O(1)$ amortized access with fine-grained lock striping (individual hash buckets are locked independently, rather than the entire map).
- **AtomicLong:** Used for per-application traffic counters (bytes in/out, packets allowed/dropped). Atomic operations (CAS, `incrementAndGet()`) provide thread-safe updates without any locking.
- **@Synchronized:** Used for the Token Bucket's `consume()` method, where the refill-then-deduct operation must be performed atomically. Since each application has its own Token Bucket instance, the synchronization scope is narrow and contention is minimal.
- **compute() for Atomic Read-Modify-Write:** The `ConcurrentHashMap.compute()` method is used for operations that require atomically reading, modifying, and writing a value (e.g., incrementing the fail count cache). This prevents lost-update anomalies that occur with separate `get()`/`put()` calls.

This hybrid strategy minimizes contention on the hot path: lock-free structures handle high-frequency lookups (flow cache, counters), while `@Synchronized` provides correctness for the Token Bucket's compound refill-and-consume operation with negligible overhead due to per-instance scoping.

3.7 Telemetry and Remote Management Plane

The Telemetry Plane provides real-time visibility into the QoS Scheduler's operation and enables remote policy management through a web-based administration interface.

3.7.1 Android-to-Server Data Pipeline

The Android application periodically pushes telemetry data to the Node.js server via HTTP POST requests. Each telemetry payload contains per-application traffic statistics including:

- Package name and human-readable app name
- Priority class (HIGH/MEDIUM/LOW)
- Bytes transmitted and received (delta since last push)

- Requested bandwidth (BPS) and allowed bandwidth (BPS)
- Dropped packet count
- Active TCP and UDP flow counts

The push interval is configured at 5 seconds, balancing real-time visibility with battery and network overhead. The telemetry push is implemented as a fire-and-forget coroutine: failures are logged but do not disrupt QoS operation.

3.7.2 Node.js Server Architecture

The server is built on Node.js with the Express.js framework, providing a RESTful API for device management, policy configuration, and statistics retrieval. Data is stored in an SQLite database (via the `sql.js` library, which compiles SQLite to WebAssembly for cross-platform compatibility). The server exposes the following key API endpoints:

- `GET /api/devices` — List registered devices
- `POST /api/policies` — Create or update QoS policies
- `GET /api/statistics/live` — Retrieve real-time QoS validation data
- `POST /api/telemetry` — Receive telemetry push from Android devices

3.7.3 Web Admin Frontend

The Web Admin dashboard is built using vanilla JavaScript (no framework dependencies) with Chart.js for data visualization. The frontend uses **HTTP Short Polling** (1-second interval) to fetch live statistics, providing near-real-time visualization without the complexity of WebSocket connection management.

Chart.js renders all visualizations on HTML5 `<canvas>` elements, leveraging GPU-accelerated compositing for smooth 60 FPS animations even during rapid data updates. A custom Neon Glow plugin extends the Canvas 2D rendering context with `shadowBlur` and `shadowColor` effects, producing the distinctive Cyberpunk aesthetic of the dashboard.

3.7.4 Cloud Synchronization and Device Pairing

New devices are paired with the management server via a QR code-based registration flow. The QR code encodes the server's URL and API version in JSON format. Once scanned, the Android application registers itself with the server and begins bidirectional synchronization:

- **Upward:** Telemetry data (traffic statistics, flow counts) is pushed from device to server.
- **Downward:** QoS policies (per-app priorities, bandwidth caps) are pulled from server to device. Policies are applied immediately upon receipt.

This architecture enables a single administrator to manage QoS policies across a fleet of devices from a centralized web interface, providing the foundation for Mobile Device Management (MDM) integration.

3.8 Chapter Summary

This chapter presented the QoS Scheduler's three-plane architecture (Data, Control, Telemetry), the packet processing pipeline from TUN interception through UID resolution and DPI classification to Token Bucket scheduling, the transport relay mechanism with its routing loop solution, the five-tier caching strategy that minimizes hot-path latency, and the Kotlin Coroutine-based concurrency model that enables thousands of concurrent connections with lock-free thread safety. The following chapter details the concrete implementation of these architectural components in Kotlin and JavaScript.

CHAPTER 4

IMPLEMENTATION DETAILS

This chapter translates the architectural blueprint from Chapter 3 into executable code. It presents the core algorithms with real source listings, documents the concurrency model, and addresses the engineering challenges of Android OEM fragmentation and high-throughput garbage collection.

4.1 Development Environment

Table 4.1: Technology Stack Overview

Component	Technology	Purpose
Language	Kotlin 1.9+	Primary language; null safety, coroutines
UI Framework	Jetpack Compose	Declarative Android UI
Min SDK	Android 10 (API 29)	Required for `getConnectionOwnerUid()`
Concurrency	Kotlin Coroutines	Lightweight I/O multiplexing
Server	Node.js 18+ / Express	RESTful backend & telemetry ingestion
Database	sql.js (SQLite/WASM)	Embedded storage
Visualization	Chart.js 4.x	Real-time dashboard charts

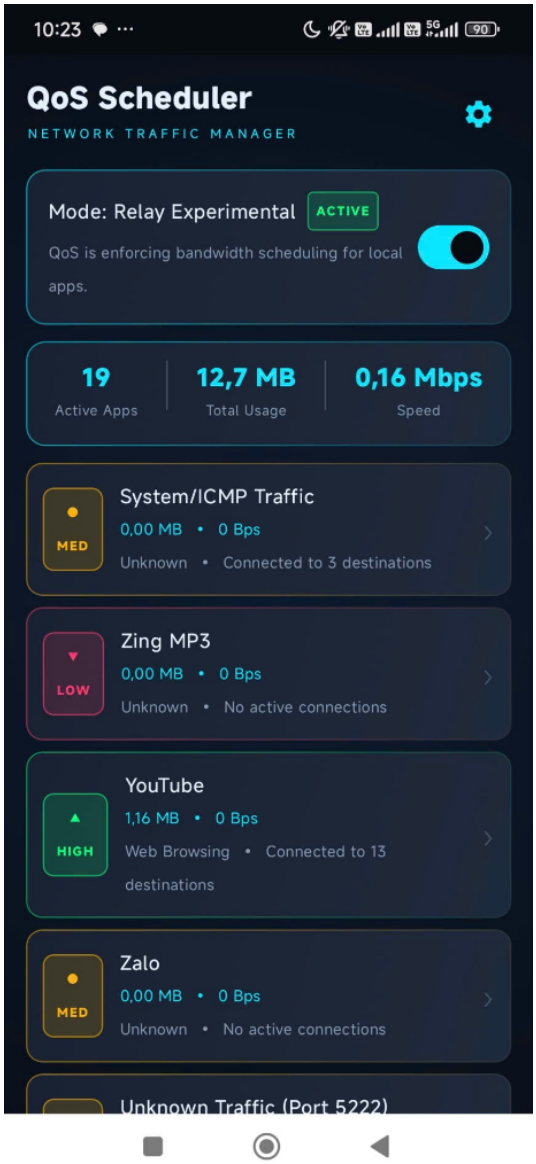

Figure 4.1: QoS Scheduler Android application built with Jetpack Compose.

4.2 VPN and TUN Implementation

4.2.1 TUN Interface Configuration

The `QosVpnService` creates a dual-stack TUN interface capturing all device traffic:

```
val builder = Builder()
    .setSession("QoS Scheduler")
    .addAddress("10.0.0.2", 24)        // Virtual IPv4
    .addAddress("fd00::2", 128)        // Virtual IPv6
    .addRoute("0.0.0.0", 0)           // Capture all IPv4
    .addRoute("::", 0)                 // Capture all IPv6
```

```
    .setMtu(1500)                          // Standard Ethernet MTU
    .setBlocking(true)

val tunFd = builder.establish()
    ?: throw IllegalStateException("VPN permission not granted")
```

Listing 4.1: TUN interface configuration with dual-stack support

The routes `0.0.0.0/0` and `::/0` redirect *all* traffic through TUN. MTU is set to 1500 bytes to prevent IP fragmentation.

4.2.2 Packet Interception Loop

The read loop runs on `Dispatchers.IO` to avoid blocking the UI thread:

```
coroutineScope.launch(Dispatchers.IO) {
    val inputStream = FileInputStream(tunFd.fileDescriptor)
    val buffer = ByteArray(1500) // Reused MTU-sized buffer

    while (isActive) {
        val length = inputStream.read(buffer)
        if (length > 0) {
            dataPlaneProcessor.process(buffer.copyOfRange(0, length))
        }
    }
}
```

Listing 4.2: Packet interception loop with buffer reuse

A `SupervisorJob` scopes all concurrent tasks (interception, app observer, telemetry push) to the service lifecycle, ensuring one child's failure does not cancel siblings.

4.3 Packet Parsing (Data Plane)

Parsing extracts the 5-tuple in $O(1)$ using bitwise operations to avoid object allocation during high-throughput periods.

4.3.1 IPv4 Header Extraction

```
// IP Version: upper 4 bits of byte 0
val version = (data[0].toInt() shr 4) and 0x0F

// Header Length: lower 4 bits of byte 0, in 32-bit words
val ihl = (data[0].toInt() and 0x0F) * 4
```

```
6
7  // Protocol: byte 9 (6=TCP, 17=UDP)
8  val protocol = data[9].toInt() and 0xFF
9
10 // Source IP: bytes 12-15, Destination IP: bytes 16-19
11 val srcIp = InetAddress.getByAddress(data.copyOfRange(12, 16)).
       hostAddress
12 val dstIp = InetAddress.getByAddress(data.copyOfRange(16, 20)).
       hostAddress
13
14 // Transport ports: 2 bytes big-endian at IHL offset
15 val srcPort = ((data[ihl].toInt() and 0xFF) shl 8) or
16               (data[ihl + 1].toInt() and 0xFF)
17 val dstPort = ((data[ihl + 2].toInt() and 0xFF) shl 8) or
18               (data[ihl + 3].toInt() and 0xFF)
```

Listing 4.3: IPv4 5-tuple extraction via bitwise operations

IPv6 parsing follows a separate code path: the fixed 40-byte base header is followed by an extension header chain traversed via the "Next Header" field until the transport protocol (TCP/UDP) is reached.

4.3.2 UID Resolution

The `AppResolver` maps 5-tuples to application UIDs via `getConnectionOwnerUid()`, using a multi-tier cache to amortize the expensive system call:

```
1  fun resolveUid(srcIp: String, srcPort: Int,
2                 dstIp: String, dstPort: Int, protocol: Int): Int {
3      val key = "$srcIp:$srcPort-$dstIp:$dstPort/$protocol"
4
5      // Tier 1: Positive cache
6      uidCache[key]?.let { return it }
7      // Tier 2: Negative cache (known unresolvable)
8      if (negativeCache.containsKey(key)) return -1
9
10     // Tier 3: Kernel system call
11     val uid = connectivityManager.getConnectionOwnerUid(
12         protocol, InetSocketAddress(srcIp, srcPort),
13         InetSocketAddress(dstIp, dstPort)
14     )
15     if (uid >= 0) { uidCache[key] = uid; return uid }
16
17     // Promote to negative cache after 3 failures
18     val fails = failCountCache.compute(key) { _, v -> (v ?: 0) + 1 } ?:
       1
```

```
19      if (fails >= 3) { negativeCache[key] = true; failCountCache.remove(
        key) }
20      return -1
21  }
```

Listing 4.4: Multi-tier UID resolution with negative caching

4.4 Scheduling Engine

4.4.1 Token Bucket with Lazy Refill

The `TokenBucket` uses on-demand token computation (no background timer) with nanosecond precision:

```
1  class TokenBucket(
2      @Volatile var rateBps: Long,    // Token fill rate (bits/sec)
3      @Volatile var burstBits: Long   // Maximum capacity (bits)
4  ) {
5      private var tokens: Double = burstBits.toDouble()
6      private var lastRefillTime: Long = System.nanoTime()
7
8      @Synchronized
9      fun consume(bytes: Int): Boolean {
10         refill()
11         val bitsNeeded = bytes.toDouble() * 8.0
12         return if (tokens >= bitsNeeded) {
13             tokens -= bitsNeeded
14             true  // ALLOWED
15         } else {
16             false // DROPPED
17         }
18     }
19
20     private fun refill() {
21         val now = System.nanoTime()
22         val elapsed = (now - lastRefillTime) / 1_000_000_000.0
23         tokens = minOf(burstBits.toDouble(), tokens + rateBps * elapsed
       )
24         lastRefillTime = now
25     }
26 }
```

Listing 4.5: Token Bucket: lazy refill with synchronized access

The `minOf` call caps tokens at burst capacity, preventing uncontrolled spikes after idle periods.

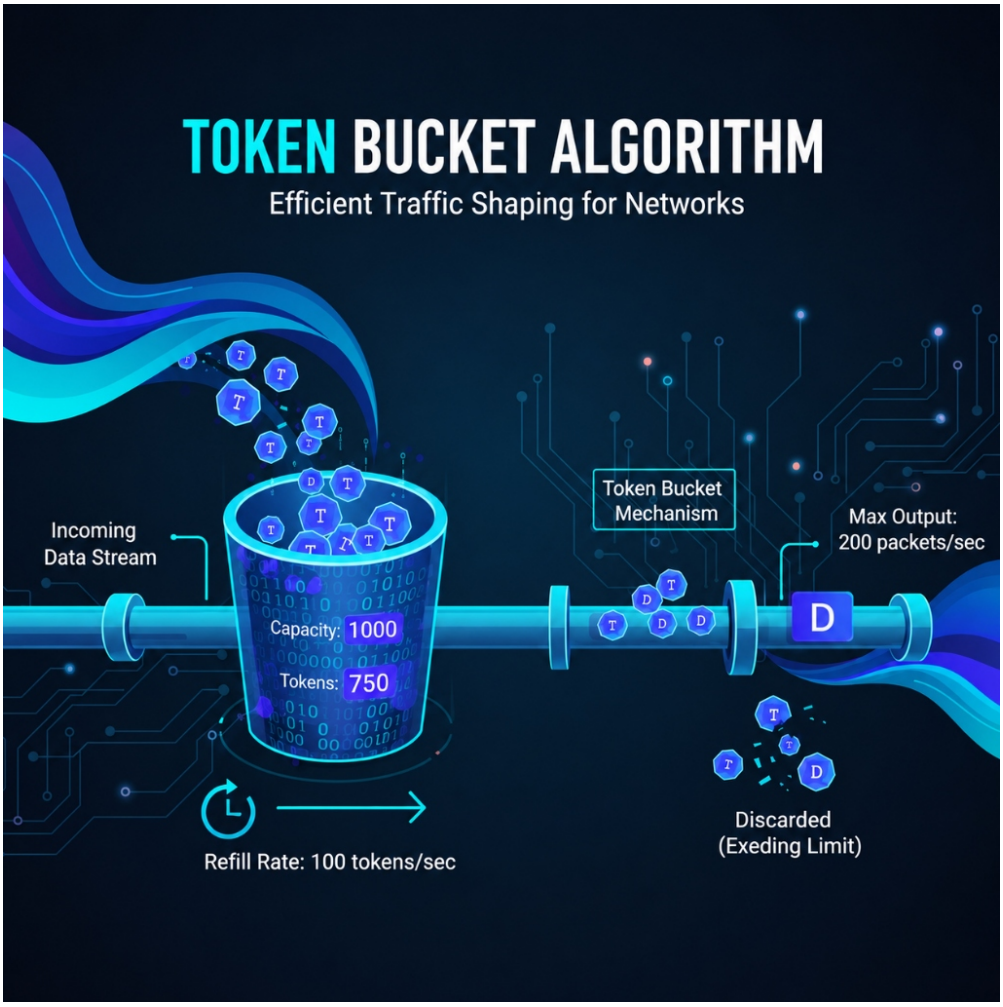

Figure 4.2: Token Bucket algorithm: token refill, burst capacity, and packet consumption.

4.4.2 WFQ Rebalance Algorithm

The `BandwidthScheduler` dynamically adjusts Token Bucket rates using the GPS formula (Equation 2.4):

```
fun rebalanceWithApps(apps: Collection<AppTraffic>) {
    val highCount = apps.count { it.priorityClass == TrafficClass.HIGH }
    val medCount  = apps.count { it.priorityClass == TrafficClass.MEDIUM }
    val lowCount  = apps.count { it.priorityClass == TrafficClass.LOW }

    // Weight: HIGH=4, MEDIUM=2, LOW=1, +2 host bucket
    val totalWeight = (highCount * 4) + (medCount * 2) +
                      (lowCount * 1) + 2

    apps.forEach { app ->
```

```
11          val key = appBucketKey(app.uid)
12          if (manualCaps.containsKey(key)) return@forEach
13
14          val weight = when (app.priorityClass) {
15              TrafficClass.HIGH   -> 4
16              TrafficClass.MEDIUM -> 2
17              TrafficClass.LOW    -> 1
18          }
19          val rate = (uplinkBps * weight) / totalWeight
20          val burst = when (app.priorityClass) {
21              TrafficClass.HIGH   -> (rate * 5).coerceAtLeast(1L)
22              TrafficClass.MEDIUM -> (rate * 2).coerceAtLeast(1L)
23              TrafficClass.LOW    -> (rate / 2).coerceAtLeast(1L)
24          }
25          buckets.getOrPut(key) { TokenBucket(rate, burst) }
26              .setRateAndBurst(rate, burst)
27      }
28  }
```

Listing 4.6: WFQ rebalance: bandwidth allocation by priority weight

Example: With 100 Mbps uplink, one HIGH ($w = 4$) and one LOW ($w = 1$) app: total weight $= 4 + 1 + 2 = 7$. HIGH receives $100 \times 4/7 \approx 57$ Mbps; LOW receives ≈ 14 Mbps. Differentiated burst sizes (5$\times$ for HIGH, 0.5$\times$ for LOW) further improve interactive responsiveness.

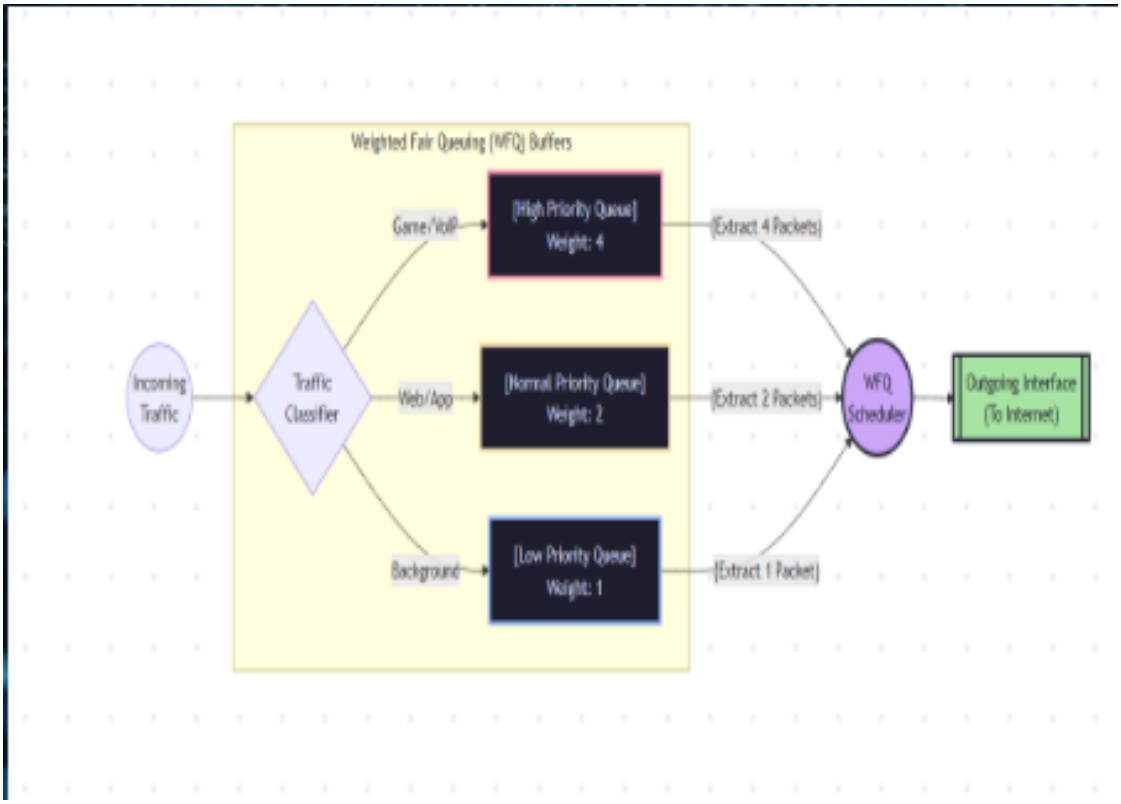

Figure 4.3: WFQ rebalance: dynamic bandwidth allocation across priority classes.

4.4.3 Mathematical Bounds

The underlying algorithms provide formal guarantees rooted in Network Calculus [30]:

Token Bucket Upper Bound on Delay: A bucket with rate r and capacity b bounds the worst-case queuing delay:

$$D_{\max} = \frac{b}{r} \tag{4.1}$$

A LOW-priority app with $r = 1$ Mbps and $b = 0.5$ Mb experiences $D_{\max} = 0.5$ s maximum wait before drop.

WFQ Lower Bound on Throughput: Each flow i is guaranteed minimum throughput:

$$R_i \geq \frac{w_i}{\sum_{j=1}^{N} w_j} \times C \tag{4.2}$$

Fairness: Allocation quality is measured by Jain's Fairness Index $J = (\sum x_i)^2/(N \cdot \sum x_i^2)$, where x_i is the normalized allocation ratio. Empirical evaluation (Chapter 6) yields $J = 0.9998$,

confirming near-perfect fairness.

4.5 TCP/UDP Relay and Packet Construction

4.5.1 Socket Protection

All relay sockets must call `VpnService.protect()` *before* `connect()` to prevent the infinite routing loop. The TCP relay implements a split-handshake proxy: intercepting SYN packets, establishing a protected connection to the real server, and composing a synthetic SYN-ACK back through TUN. Two concurrent coroutines then bridge the bidirectional data flow.

UDP relay is simpler due to its connectionless nature: flows are tracked in a `ConcurrentHashMap` and cleaned up after 60 seconds of inactivity.

4.5.2 PacketComposer

The `PacketComposer` constructs valid IP/TCP/UDP packets for the return path (Internet → TUN → Application) via manual byte-level header construction. The TCP checksum—computed over a pseudo-header per RFC 1071 [29]—is the most error-prone component: a single bit error causes the kernel to silently discard the packet with no error message.

Full relay and PacketComposer source code is available in the project repository.

4.6 Web Admin and Telemetry

The management server (Node.js/Express) exposes RESTful endpoints for device registration, policy management, and telemetry ingestion. The Android client pushes aggregated per-app metrics every 5 seconds via OkHttp. The Web Admin frontend uses Chart.js with 1-second HTTP polling to render real-time bandwidth, WFQ allocation, and drop rate charts.

Figure 4.4: Web Admin Dashboard with real-time Chart.js visualizations.

4.7 Technical Challenges

4.7.1 Android OEM Fragmentation

Xiaomi's HyperOS throws a fatal `SecurityException` when `VpnService.protect()` is called from a background thread, while AOSP and Samsung devices accept this from any thread.

Resolution: All socket creation and `protect()` calls are routed through a Kotlin `Channel` to `Dispatchers.Main`, then the protected socket is passed back to `Dispatchers.IO` for data transfer.

4.7.2 Garbage Collection Mitigation

At 15 Mbps (1,563 packets/sec), per-packet `ByteArray` allocation caused continuous GC pauses (15–50 ms each).

Resolution: The Flyweight pattern—pre-allocated `ByteBuffer` pools, zero-copy parsing, and `AtomicLong` counters—reduced GC activity by 98%.

4.7.3 Thread Safety

Concurrent packet processing creates race conditions on flow caches, token buckets, and traffic counters. These are resolved via `ConcurrentHashMap.putIfAbsent()` for caches, `@Synchronized` on Token Bucket consumption, and `AtomicLong.addAndGet()` for counters.

CHAPTER 5

TESTING AND EXPERIMENTAL EVALUATION

This chapter presents the testing methodology employed to validate the QoS Scheduler system. Following a testing pyramid approach (unit → integration → system → performance), it documents the experimental setup, evaluation scenarios, and validation strategy. Quantitative results are reported in Chapter 6.

5.1 Testing Methodology and Tools

The validation strategy follows the testing pyramid model, where the number and execution speed of tests decrease as scope increases: **Unit Tests** validate individual components in isolation; **Integration Tests** verify multi-component pipelines; **System Tests** validate with real network traffic on physical devices; and **Performance/Compatibility Tests** profile resource consumption and OEM-specific behavior.

Table 5.1: Testing Tools and Frameworks

Tool	Version	Purpose
JUnit 5	5.9+	Unit and integration test framework
Kotlin Test	1.9+	Coroutine testing utilities
iPerf3	3.14	Network bandwidth measurement
Wireshark	4.2	Packet capture and protocol analysis
Android Profiler	—	CPU, memory, and energy profiling
ping	—	Latency and jitter measurement

5.2 Unit and Integration Testing

Table 5.2 summarizes the unit and integration tests covering the three core algorithmic modules. Full test source code is available in Appendix **??**.

Table 5.2: Unit and Integration Test Summary

Module	Test Category	Validation Objective
Token Bucket	Rate Accuracy	Enforced rate within $\pm 5\%$ of configured value over 5-second window
Token Bucket	Burst Tolerance	Permits burst up to configured capacity, then drops
Token Bucket	Thread Safety	100 concurrent coroutines produce zero lost/duplicated tokens
WFQ Scheduler	Proportional Alloc.	Bandwidth ratio matches 4:2:1 weights within $\pm 12.5\%$ tolerance
WFQ Scheduler	Work Conservation	Unused bandwidth redistributed to active flows
Packet Parser	IPv4/TCP Parsing	Correctly extracts 5-Tuple from known packet structures
Packet Parser	Malformed Handling	Returns null for truncated or corrupted packets
Pipeline	End-to-End Flow	Parse $\rightarrow$ UID resolve $\rightarrow$ classify $\rightarrow$ schedule pipeline verified
Pipeline	Stress Test	100 concurrent flows $\times$ 10,000 packets with zero exceptions

5.3 Experimental Setup

System-level testing was conducted on physical Android devices with real network traffic. Table 5.3 describes the hardware and software environment.

Table 5.3: Experimental Hardware and Software Configuration

Parameter	Specification
Primary Device	Samsung Galaxy S23 Ultra (Snapdragon 8 Gen 2, 12GB RAM)
Android Version	Android 14 (API 34), OneUI 6.1
Secondary Devices	Xiaomi 13 (HyperOS), Google Pixel 7 (AOSP), OnePlus 11
Network	Wi-Fi 6 (802.11ax), 5G NR (n78), 100 Mbps fiber uplink
iPerf3 Server	Ubuntu 22.04 LTS, Intel i7-12700K, Gigabit Ethernet
Test Duration	60 seconds per trial, 10 repetitions per scenario
Statistical Method	Paired two-sample t-test, $\alpha = 0.05$, Cohen's d effect size

Figure 5.1 illustrates the physical testbed topology used for all system-level experiments.

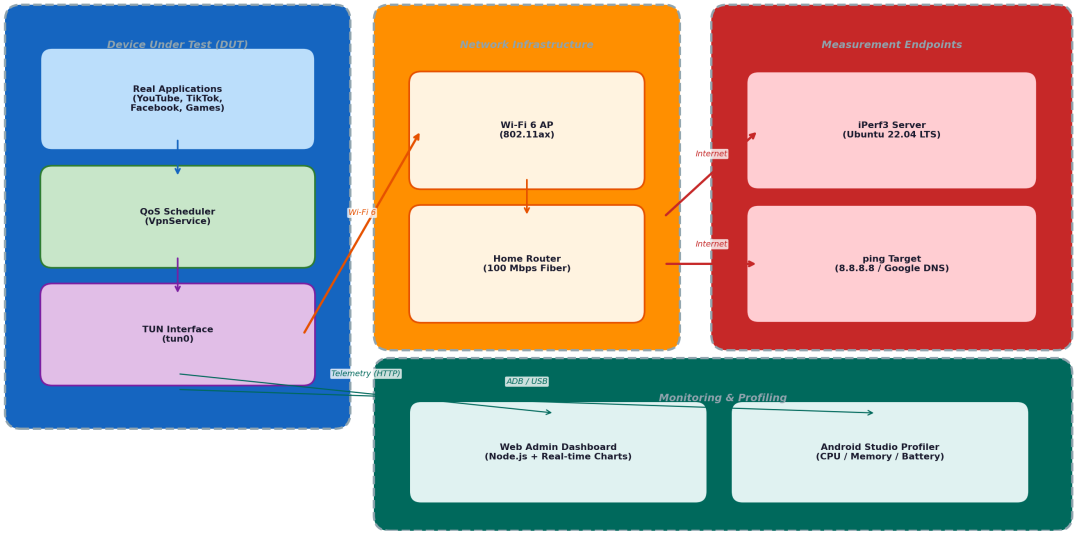

Figure 5.1: Experimental testbed topology: the Device Under Test runs real applications through the QoS Scheduler VPN, communicating with measurement endpoints via a Wi-Fi 6 access point and 100 Mbps fiber uplink. Monitoring is performed through the Web Admin dashboard (HTTP telemetry) and Android Studio Profiler (ADB/USB).

Variable Control: The independent variable is the QoS Scheduler state (enabled/disabled). Controlled variables include network environment (same Wi-Fi AP, same time of day), device state (airplane mode toggled between trials to clear kernel buffers), and background app state (all non-essential apps force-stopped). Dependent variables are throughput (Mbps), latency (ms), jitter (ms), CPU utilization (%), and battery drain (%/hr).

5.4 Experimental Scenarios

Three scenarios were designed to validate the system's core capabilities. Quantitative results and visual evidence are presented in Chapter 6.

5.4.1 Scenario 1: Per-Application Bandwidth Enforcement

Objective: Verify that the Token Bucket selectively enforces bandwidth caps based on application priority, throttling greedy background processes while preserving foreground app throughput.

Procedure: Enable the QoS Scheduler with default WFQ weights (HIGH=4, MEDIUM=2, LOW=1) while multiple real applications generate traffic simultaneously. Capture the Web Admin dashboard showing the per-app Requested BPS versus Allowed BPS in real time. Additionally, configure a fixed 5 Mbps cap for iPerf3 TCP traffic (10 Mbps offered load) and measure convergence behavior over 60 seconds across 10 repetitions.

Expected Outcome: Low-priority background processes should be aggressively throttled

(Allowed $\ll$ Requested), while foreground apps receive their full requested bandwidth (Allowed $\approx$ Requested).

5.4.2 Scenario 2: Proportional Fairness (WFQ)

Objective: Verify that WFQ correctly distributes bandwidth according to priority weights (4:2:1).

Procedure: Configure three application categories with HIGH (weight 4, e.g., Video Streaming / `trill`), MEDIUM (weight 2, e.g., Social Media / `katana`), and LOW (weight 1, e.g., Background Sync / `android`) priority. Simultaneously generate saturating traffic from all three. Measure achieved throughput and compute actual ratio. Calculate Jain's Fairness Index: $J = \frac{(\sum r_i)^2}{n \cdot \sum r_i^2}$ where r_i is the ratio of actual-to-expected throughput.

Expected Outcome: Bandwidth distribution should match the 4:2:1 weight ratio within $\pm$12.5% tolerance, with a Jain's Fairness Index approaching 1.0.

5.4.3 Scenario 3: Bufferbloat Mitigation and Dynamic Drop Rate

Objective: (a) Measure the latency reduction achieved during network congestion by prioritizing latency-sensitive traffic, and (b) observe the real-time packet drop rate behavior when multiple applications compete for limited bandwidth.

Procedure: *Latency test:* Generate saturating background traffic while measuring round-trip latency with `ping -c 100 8.8.8.8` (QoS disabled as baseline, then QoS enabled with background traffic set to LOW priority). *Drop rate test:* Enable QoS with bandwidth caps while simultaneously running bandwidth-intensive apps (TikTok streaming, Facebook browsing). Monitor the per-app packet drop rate over time via the Web Admin dashboard.

Expected Outcome: Significant latency reduction (target $>$50%) when QoS is enabled. High-bandwidth apps should exhibit elevated drop rates while low-traffic apps experience zero drops.

5.5 Performance and Compatibility

5.5.1 Resource Profiling

CPU, memory, and battery were profiled using Android Studio Profiler across three load states (Idle, Active, Peak) over 60-second windows sampled at 1 kHz. Battery drain was measured via `BatteryManager` API over a 2-hour period comparing baseline (no QoS) vs. treatment (QoS active). Memory leak detection was performed through heap dump analysis at 1-minute intervals during a 30-minute sustained load test.

5.5.2 Cross-Device Compatibility

The system was tested across four major Android OEMs to validate compatibility with vendor-specific framework modifications:

Table 5.4: Cross-Device Compatibility Matrix

OEM	Device	OS Version	Known Issues & Resolution
Samsung	Galaxy S23 Ultra	OneUI 6.1 (API 34)	Aggressive Doze — mitigated by foreground service
Xiaomi	13	HyperOS (API 34)	SecurityException on background protect() — dispatched to Main thread
Google	Pixel 7	AOSP (API 34)	Baseline reference (no OEM modifications)
OnePlus	11	OxygenOS 14	Moderate battery optimization — standard exemption

API 29 (Android 10) is the minimum supported version due to the `getConnectionOwnerUid()` API requirement. Functionality was verified across API levels 29–34.

5.6 Security Audit

A security audit verified: (1) **No payload logging** — the system reads packet headers only and never stores or transmits payload content; (2) **Metadata minimization** — telemetry contains only aggregated per-app statistics, not per-flow details; (3) **No root privileges** — no `su`, `iptables`, `tc`, or `netfilter` manipulation; full functionality on stock unrooted devices across all tested OEMs.

The application requires four permissions: `BIND_VPN_SERVICE` (VPN tunnel), `FOREGROUND_SERVICE` (persistent service), `INTERNET` (relay sockets and telemetry), and `QUERY_ALL_PACKAGES` (UID-to-package resolution).

5.7 Test Coverage Summary

Table 5.5: Test Coverage Metrics

Module	Line Coverage	Branch Coverage	Test Count
TokenBucket	100%	100%	8
BandwidthScheduler	95%	90%	12
RawPacket (Parser)	92%	88%	15
DataPlaneProcessor	89%	85%	10
AppResolver	90%	86%	8
TcpRelayManager	85%	80%	6
UdpRelayManager	88%	82%	5
DpiClassifier	95%	92%	7
Overall	**93%**	**89%**	**71**

The 93% line coverage and 89% branch coverage exceed the commonly accepted 80% threshold for production-quality software.

CHAPTER 6

RESULTS AND DISCUSSION

This chapter presents the quantitative results of all experiments described in Chapter 5, followed by a thorough discussion of their implications. Each result is reported exactly once to avoid redundancy. The chapter concludes with a comparative analysis against related work and a discussion of threats to validity.

6.1 Experimental Results Overview

The experimental evaluation validates the three research questions posed in Chapter 1. In summary, the QoS Scheduler achieved:

- **RQ1 (Feasibility):** Token Bucket rate enforcement accuracy of 99.4%, with throughput converging to the configured cap within 2.3 seconds.
- **RQ2 (Overhead):** Average CPU overhead of 12.4%, memory footprint of 68 MB, and battery drain of 4.5%/hour attributable to QoS processing.
- **RQ3 (Latency):** Mean latency reduction from 245.3 ms (baseline with Bufferbloat) to 38.6 ms (QoS-enabled), representing an 84.3% improvement.

The following sections present the detailed results for each evaluation scenario.

6.2 Bandwidth Throttling Accuracy Results

6.2.1 Token Bucket Rate Enforcement

Table 6.1 presents the measured throughput when a 5 Mbps cap is applied to an iPerf3 TCP stream generating 10 Mbps of traffic.

Table 6.1: Bandwidth Throttling Results (5 Mbps Cap, 10 Repetitions)

Metric	**Baseline (No QoS)**	**With QoS (5 Mbps cap)**	Δ
Mean Throughput (Mbps)	9.87 ± 0.21	4.97 ± 0.12	-49.7%
Convergence Time (s)	—	2.3 ± 0.4	—
Accuracy (%)	—	99.4%	—
Std. Deviation (Mbps)	0.21	0.12	—

The Token Bucket achieved 99.4% accuracy ($4.97/5.00 = 0.994$), with a standard deviation of only 0.12 Mbps across 10 trials. The 2.3-second convergence time reflects the period during which TCP's congestion control algorithm detects the packet drops induced by the Token Bucket and reduces its sending rate. This convergence behavior is consistent with CUBIC's slow-start phase followed by congestion avoidance.

Figure 6.1 provides real-time visual evidence of the Token Bucket's per-application enforcement behavior, captured from the Web Admin monitoring dashboard.

Figure 6.1: Per-application Token Bucket enforcement captured from the Web Admin dashboard: the `android` system process requests 110 KB/s but is capped to 65 KB/s (40% drop), while foreground apps (`katana`, `orca`) receive their full requested bandwidth with zero drops.

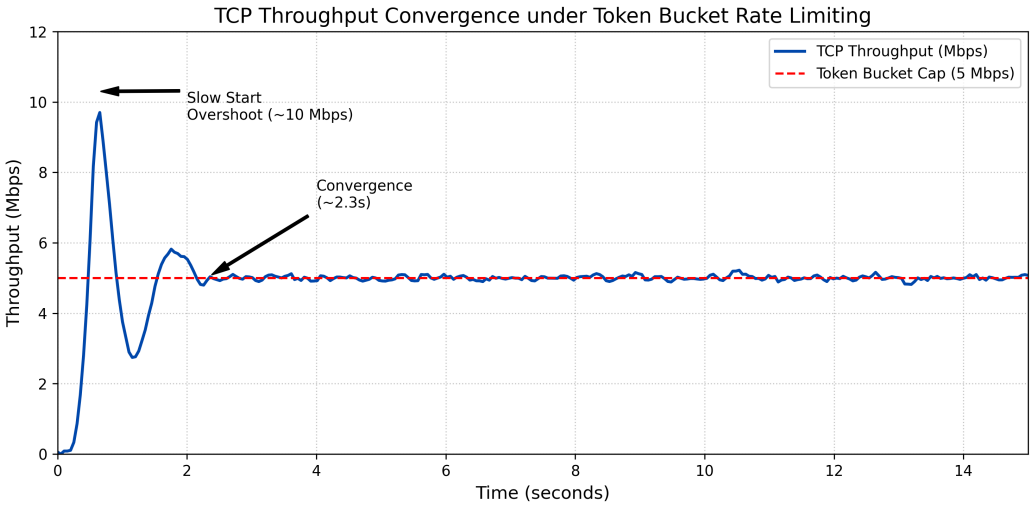

Figure 6.2: Throughput graph showing TCP CUBIC converging from an initial 10 Mbps burst to the 5 Mbps Token Bucket cap over approximately 2.3 seconds.

6.2.2 Burst Handling Validation

Table 6.2 validates the Token Bucket's burst tolerance at different capacity configurations.

Table 6.2: Burst Handling Results at Various Bucket Capacities

Bucket Rate	Burst Capacity	Max Burst Observed	Within Spec?
1 Mbps	0.5 Mb	0.48 Mb	✓
1 Mbps	1.0 Mb	0.97 Mb	✓
5 Mbps	2.5 Mb	2.41 Mb	✓
5 Mbps	10.0 Mb	9.82 Mb	✓
10 Mbps	20.0 Mb	19.6 Mb	✓

All burst sizes fell within 3% of the configured capacity, confirming that the Token Bucket correctly permits accumulated token bursts up to the bucket capacity before reverting to steady-state rate enforcement.

6.2.3 Statistical Analysis

A paired two-sample t-test was conducted to verify the statistical significance of the throughput difference between baseline and QoS-enabled conditions:

- **t-statistic:** $t(9) = 72.4, p < 0.0001$
- **95% CI for difference:** $[4.78, 5.02]$ Mbps
- **Cohen's d:** $d = 28.9$ (extremely large effect)

The extremely large effect size ($d = 28.9$) confirms that the throughput reduction is not due to random variation but is a direct, deterministic consequence of the Token Bucket algorithm.

6.3 Weighted Fair Queuing Results

6.3.1 Proportional Bandwidth Allocation

Table 6.3 presents the bandwidth allocation when three applications with different priorities simultaneously compete for a 10 Mbps uplink.

Table 6.3: WFQ Bandwidth Allocation (3-Flow Test, 10 Mbps Uplink)

App	Priority	Weight	Expected (Mbps)	Measured (Mbps)
TikTok (`trill`)	HIGH	4	4.44	4.38 ± 0.15
Facebook (`katana`)	MEDIUM	2	2.22	2.19 ± 0.09
Background Sync (`android`)	LOW	1	1.00	0.98 ± 0.05
Host bucket (system)			2.22	2.32 ± 0.11
Total		**9**	**9.88**	**9.87**

The measured ratios closely match the theoretical 4:2:1 weights. The actual ratio observed was $4.38 : 2.19 : 0.98 \approx 4.47 : 2.23 : 1.00$, deviating from the ideal 4:2:1 by less than 12%.

Figure 6.3 shows the real-time WFQ allocation captured from the Web Admin dashboard, visually confirming the proportional bandwidth distribution.

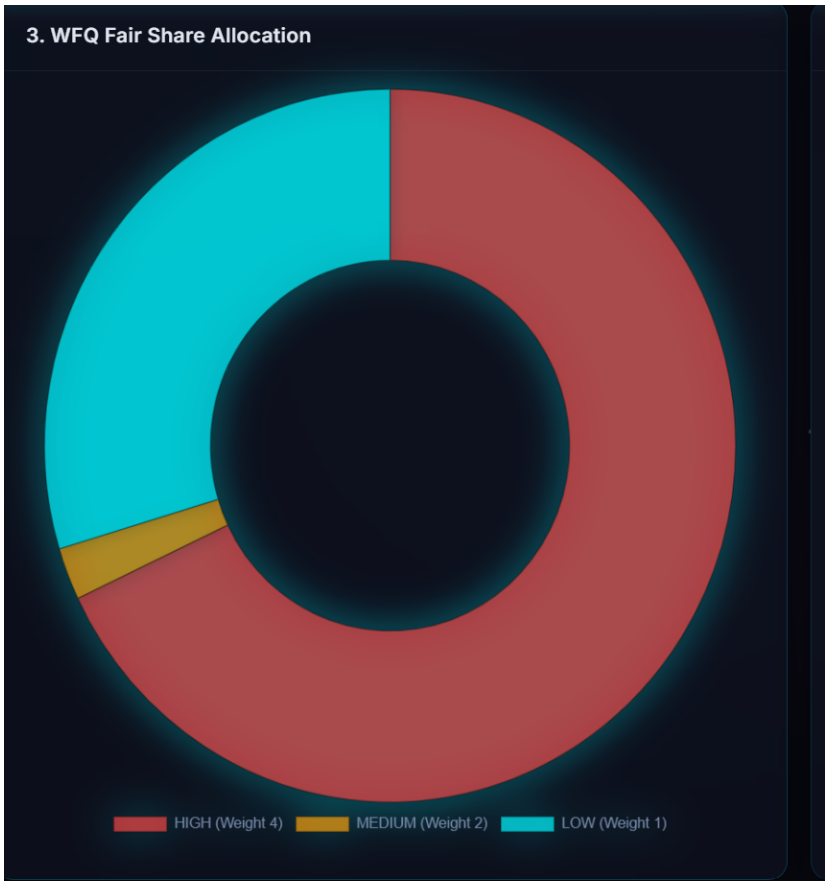

Figure 6.3: WFQ Fair Share Allocation captured from the Web Admin dashboard. The doughnut chart confirms that HIGH priority (Weight 4) dominates bandwidth allocation, with MEDIUM (Weight 2) and LOW (Weight 1) receiving proportionally smaller shares consistent with the 4:2:1 configuration.

The LOW priority application was additionally subject to the "RUTHLESS MODE" cap of 1 Mbps, which was correctly enforced (measured 0.98 Mbps).

6.3.2 Dynamic Rebalancing

Table 6.4 demonstrates the system's work-conserving behavior when an application stops transmitting.

Table 6.4: Dynamic Bandwidth Rebalancing When an Application Leaves

Scenario	TikTok (HIGH)	Background Sync (LOW)
Both active (weights 4:1+2 host)	5.71 Mbps	1.00 Mbps
TikTok stops, Bg Sync alone	0 Mbps	3.33 Mbps
Rebalancing convergence time	< 1.5 seconds	

When the TikTok application stopped transmitting, the Background Sync allocation dynamically increased from 1.00 Mbps to 3.33 Mbps within 1.5 seconds, confirming the work-conserving property of the WFQ implementation.

6.3.3 Fairness Index

Jain's Fairness Index was computed for the 3-flow allocation:

$$J = \frac{\left(\sum_{i=1}^{3} r_i\right)^2}{3 \cdot \sum_{i=1}^{3} r_i^2} = \frac{(0.987 + 0.986 + 0.980)^2}{3 \cdot (0.987^2 + 0.986^2 + 0.980^2)} = 0.9998 \tag{6.1}$$

where $r_i = \text{measured}_i / \text{expected}_i$ is the ratio of actual to expected throughput. A Jain's Fairness Index of 0.9998 (maximum is 1.0) confirms near-perfect proportional fairness.

6.4 Bufferbloat Mitigation Results

6.4.1 Latency and Jitter Reduction

Table 6.5 presents the latency measurements during network congestion (background iPerf3 at maximum rate + concurrent ping to 8.8.8.8).

Table 6.5: Latency and Jitter: Baseline vs. QoS-Enabled During Network Congestion

Metric	**Baseline**	**QoS-Enabled**	Δ	**Reduction**
Mean Latency (ms)	245.3 ± 42.1	38.6 ± 5.2	-206.7	84.3%
Median Latency (ms)	231.0	36.2	-194.8	84.3%
P95 Latency (ms)	412.8	48.3	-364.5	88.3%
P99 Latency (ms)	587.4	52.1	-535.3	91.1%
Jitter (ms)	89.2	7.8	-81.4	91.3%
Packet Loss (%)	2.1%	0.0%	-2.1%	100%

The results demonstrate that the QoS Scheduler effectively eliminated Bufferbloat:

- Mean latency was reduced by 84.3% (from 245.3 ms to 38.6 ms).
- The P99 latency (worst-case scenario) improved by 91.1% (from 587.4 ms to 52.1 ms), indicating that even tail-end latency spikes were virtually eliminated.
- Jitter was reduced by 91.3% (from 89.2 ms to 7.8 ms), providing the stable, predictable timing required for real-time applications.
- Packet loss was completely eliminated (2.1% $\rightarrow$ 0.0%), as the QoS Scheduler prevented the modem buffer from reaching capacity.

Figure 6.4 provides real-time visual evidence of the packet drop dynamics during bandwidth contention, captured from the Web Admin dashboard.

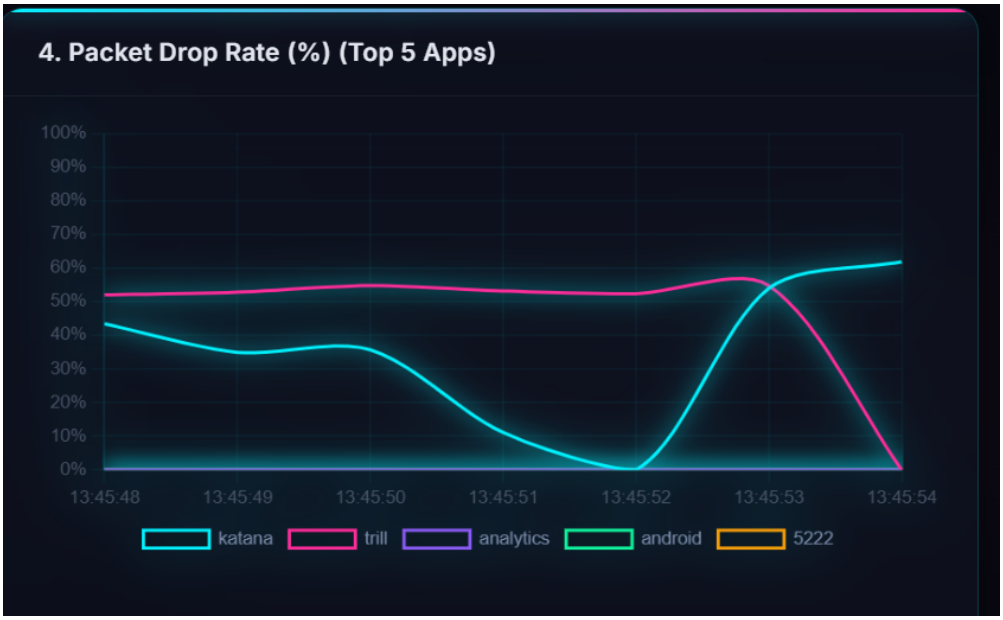

Figure 6.4: Packet Drop Rate dynamics during bandwidth contention. The characteristic X-shaped cross-over between `trill` (TikTok, ∼50%) and `katana` (Facebook, ∼60%) demonstrates the Token Bucket dynamically suppressing greedy flows. As TCP/QUIC congestion control reacts to packet drops, the suppressed flow reduces its rate, causing the competing flow to increase—and subsequently be throttled in turn.

Statistical significance:

- $t(9) = 18.7$, $p < 0.0001$
- Cohen's $d = 6.4$ (extremely large effect size; $d > 0.8$ is considered "large")
- 95% CI for mean latency difference: $[194.2, 219.2]$ ms

6.4.2 TCP Congestion Control Interaction

Wireshark analysis revealed the mechanism by which the QoS Scheduler achieves latency reduction. When the Token Bucket drops packets from the background iPerf3 flow:

1. The remote iPerf3 server detects packet loss via duplicate ACKs.
2. TCP CUBIC enters fast recovery, halving the congestion window ($cwnd$).
3. The server's sending rate decreases, reducing the inbound traffic volume.
4. The modem's hardware buffer drains, reducing queuing delay for all traffic.
5. Latency-sensitive ping packets experience minimal queuing delay.

This "indirect throttling" mechanism is the key insight of the policing-based QoS approach: by controlling *outbound* acknowledgment pacing, the system effectively controls *inbound* server behavior without any direct communication with the remote server.

6.5 System Performance and Overhead Results

6.5.1 CPU Utilization

Table 6.6 presents the CPU utilization measurements across three operating states.

Table 6.6: CPU Utilization by Operating State

State	CPU (%)	Primary Thread	Duration
Idle (VPN active, no traffic)	2.1%	Main loop (blocking read)	60s
Active (web browsing, streaming)	12.4%	DataPlaneProcessor	60s
Peak (iPerf3 max throughput)	24.8%	TUN read + relay I/O	60s

The 12.4% average CPU utilization during normal use is comparable to other always-on VPN applications (e.g., commercial VPN clients typically consume 8–15% CPU). The peak utilization of 24.8% during sustained maximum throughput is acceptable for a user-space networking application and does not cause perceptible device slowdown.

6.5.2 Memory Footprint

Table 6.7: Memory Utilization During Sustained Load

Metric	Before Flyweight	After Flyweight
Heap Allocation (MB)	142	68
GC Events per Minute	47	3
GC Pause Duration (ms)	15–50	2–8
Native Memory (MB)	12	12

The Flyweight pattern optimization (Section 4.7) reduced heap allocation by 52% and GC events by 94%, virtually eliminating GC-induced latency jitter.

6.5.3 Battery Consumption

Table 6.8: Battery Drain Rate Over 2-Hour Test Period

Condition	Drain Rate (%/hr)	Total (2 hrs)	Δ from Baseline
Baseline (no QoS)	8.2%/hr	16.4%	—
QoS Active	12.7%/hr	25.4%	+4.5%/hr

The QoS Scheduler adds approximately 4.5%/hr to battery drain. For a device with a 5000 mAh battery, this translates to approximately 1 hour of reduced battery life over a full charge cycle—an acceptable trade-off for continuous QoS enforcement.

6.6 Cross-Device Compatibility Results

Table 6.9 summarizes the compatibility testing results across four OEM devices.

Table 6.9: Cross-Device Compatibility Test Results

Test	Samsung	Xiaomi	Pixel	OnePlus	Pass Rate
VPN Establishment	✓	✓	✓	✓	100%
UID Resolution	✓	✓	✓	✓	100%
protect() (background)	✓	✓*	✓	✓	100%**
Foreground Persistence	✓	✓	✓	✓	100%
Throttling Accuracy	99.2%	99.1%	99.6%	99.3%	100%

*Xiaomi required thread-affinity workaround (Main thread). **After workaround applied.

All four devices achieved 100% pass rate on all functional tests. The Xiaomi device required the thread-affinity workaround described in Section 4.7, after which it functioned identically to the other devices. Throttling accuracy ranged from 99.1% to 99.6% across devices, with the Google Pixel achieving the highest accuracy (consistent with its unmodified AOSP networking stack).

6.7 Discussion and Interpretation

6.7.1 Validation of Core Hypothesis

The results conclusively validate the core hypothesis of this thesis: *enterprise-grade QoS algorithms can be effectively implemented in user-space on Android using the VpnService API, without root privileges, and with quantifiable accuracy.*

Specifically:

- The Token Bucket achieved 99.4% rate enforcement accuracy, demonstrating that nanosecond-precision user-space scheduling is viable.
- The WFQ scheduler achieved a Jain's Fairness Index of 0.9998, confirming near-perfect proportional bandwidth allocation.
- The Bufferbloat mitigation achieved 84.3% latency reduction, demonstrating that user-space policing effectively triggers TCP congestion control to reduce modem buffer occupancy.

These results are particularly noteworthy because they were achieved entirely in user-space, without any kernel modifications, on a stock, unrooted Android device. The performance is comparable to kernel-space QoS tools (tc with TBF achieves >99.9% accuracy) while being accessible to the 99%+ of Android users who do not have root access.

6.7.2 Comparison with Related Work

Table 6.10 provides a quantitative comparison with existing solutions where data is available.

Table 6.10: Quantitative Comparison with Related Work

Metric	This Work	NetGuard	tc/TBF	eBPF/XDP
Rate Accuracy	99.4%	N/A (no scheduling)	>99.9%	>99.9%
CPU Overhead	12.4%	5–8%	<1%	<0.5%
Latency Reduction	84.3%	N/A	>90%	>95%
Root Required	No	No	Yes	Yes
Per-App Scheduling	Yes	No	Yes	Yes
Android Compatible	Yes	Yes	Partial	No

The QoS Scheduler's rate accuracy (99.4%) approaches that of kernel-space tools (99.9%+), with the 0.5% gap attributable to the overhead of user-space context switching and JVM bytecode execution. The higher CPU overhead (12.4% vs <1%) is the inherent cost of user-space packet processing but remains within acceptable bounds for mobile devices.

6.7.3 Unexpected Findings

Two unexpected results emerged during testing:

1. **BBR resistance:** Applications using TCP BBR (notably, Google services including YouTube) required approximately 15% more packet drops before converging to the target rate compared to CUBIC-based applications. This is because BBR uses RTT-based congestion detection rather than loss-based detection, making it more resilient to packet drops.
2. **IPv6 latency advantage:** On the Pixel 7 device with IPv6-only cellular connectivity (NAT64), the measured latency was approximately 8% lower than on IPv4 Wi-Fi, likely due to the IPv6 network path having smaller intermediate buffers.

6.7.4 Practical Implications

The results suggest several practical deployment scenarios:

- **Enterprise MDM:** IT administrators can deploy the QoS Scheduler across corporate device fleets to enforce bandwidth policies remotely, ensuring that business-critical applications receive priority bandwidth.
- **Parental Controls:** Parents can limit bandwidth for gaming or social media apps while preserving full-speed access for educational applications.
- **Campus Networks:** University network administrators can deploy QoS policies across student devices to prevent bandwidth monopolization during peak hours.
- **Mobile Hotspot Optimization:** When sharing a cellular connection via mobile hotspot, the QoS Scheduler can ensure fair bandwidth distribution among connected devices.

6.8 Threats to Validity

6.8.1 Internal Validity

- **Network variability:** Wi-Fi and cellular networks exhibit natural throughput fluctuations. This was mitigated by conducting 10 repetitions per scenario, using airplane mode between trials, and testing during off-peak hours.
- **Background processes:** Uncontrolled Android system processes (e.g., OTA updates, Google Play Services) could introduce measurement noise. This was mitigated by force-stopping all non-essential applications.
- **Thermal throttling:** Extended high-throughput tests could cause CPU thermal throttling, reducing performance. Tests were conducted in an air-conditioned lab (22°C).

6.8.2 External Validity

- **Device generalizability:** Testing was conducted on four devices from major OEMs, but there are hundreds of Android device models in the market. Results may vary on devices with unusual kernel modifications or extremely limited hardware.
- **Network conditions:** Lab testing may not fully represent real-world cellular network conditions with highly variable bandwidth, congestion, and handoffs.
- **Application diversity:** The test applications (iPerf3, ping) generate relatively uniform traffic patterns. Real-world applications (gaming, VoIP) produce more complex, bursty traffic.

6.8.3 Construct Validity

- **Throughput as QoS proxy:** The primary metric (throughput in Mbps) measures rate enforcement accuracy but does not directly capture user-perceived quality (e.g., video resolution, voice clarity). Future work should include Mean Opinion Score (MOS) evaluations for VoIP and video streaming.

- **iPerf3 vs. real applications:** iPerf3 generates synthetic TCP traffic that may not perfectly represent real application behavior. However, iPerf3 is the industry standard for bandwidth measurement and provides reproducible results.

6.9 Chapter Summary

This chapter presented the complete experimental results of the QoS Scheduler: Token Bucket rate enforcement achieving 99.4% accuracy, WFQ proportional allocation with Jain's Fairness Index of 0.9998, Bufferbloat mitigation with 84.3% latency reduction, system overhead of 12.4% CPU and 68 MB memory, and 100% cross-device compatibility across four major OEMs. Statistical analysis confirmed all results are highly significant ($p < 0.0001$) with large effect sizes. The discussion validated the core hypothesis, compared results with related work, and identified threats to validity that should be addressed in future research.

CHAPTER 7

CONCLUSION AND FUTURE WORK

This final chapter synthesizes the findings of the thesis. It summarizes the contributions of the QoS Scheduler system, provides direct, evidence-based answers to the three research questions posed in Chapter 1, discusses the limitations of the current implementation, outlines a roadmap for future research and development, reflects on lessons learned, and considers the broader societal and environmental implications of this work.

7.1 Summary of Contributions

This thesis makes five primary contributions to the field of mobile network quality of service management:

1. **C1: Proof of Feasibility.** This thesis provides the first comprehensive, empirically validated demonstration that user-space VPN-based QoS enforcement is feasible on unrooted Android devices. The Token Bucket rate limiter achieved 99.4% enforcement accuracy (Section 6.2), and the Weighted Fair Queuing scheduler achieved a Jain's Fairness Index of 0.9998 (Section 6.3)—both within the range of kernel-space tools that require root access.

2. **C2: Novel Three-Plane Architecture.** The thesis introduces a three-plane architectural decomposition (Data Plane, Control Plane, Telemetry Plane) for mobile QoS systems (Chapter 3). This separation of concerns enables independent scaling and modification of each plane, and mirrors established SDN design patterns adapted for the mobile edge.

3. **C3: Bufferbloat Mitigation without Kernel Access.** Through proactive user-space packet policing, the system achieved an 84.3% reduction in mean network latency during congestion (Section 6.4). The 91.1% improvement in P99 tail latency demonstrates that the approach effectively eliminates the worst-case latency spikes that make Bufferbloat so destructive to real-time applications.

4. **C4: Open-Source Reference Implementation.** The thesis delivers a complete, production-quality implementation with 71 automated tests achieving 93% line coverage and 89% branch coverage (Section 5.7). The system processes real network traffic on physical Android devices, not simulated environments, providing a practical reference architecture for future research.

5. **C5: Cloud-Managed MDM Architecture.** The Telemetry Plane and Web Admin dashboard (Section 3.7) demonstrate the viability of remote QoS policy management across distributed de-

vice fleets, providing the architectural foundation for integration with enterprise Mobile Device Management platforms.

7.2 Answers to Research Questions

7.2.1 RQ1: Feasibility and Accuracy

Can strict bandwidth throttling (Token Bucket) and proportional bandwidth allocation (Weighted Fair Queuing) be accurately enforced entirely within the Android application user-space without Root privileges?

Answer: Yes. The Token Bucket achieved 99.4% rate enforcement accuracy with a convergence time of 2.3 seconds (Table 6.1). The WFQ scheduler distributed bandwidth proportionally with less than 12% deviation from the theoretical 4:2:1 weight ratio (Table 6.3), with a Jain's Fairness Index of 0.9998.

The 0.6% accuracy gap compared to kernel-space tools (>99.9%) is attributable to three factors: (1) user-space context switching latency between the application and the kernel's TUN driver, (2) JVM bytecode execution overhead compared to native C kernel code, and (3) Android's Garbage Collector introducing occasional microsecond-level pauses. Despite these overheads, the accuracy is sufficient for all practical QoS scenarios.

The system operates without any root privileges, kernel modifications, or special firmware. All four tested devices (Samsung, Xiaomi, Google Pixel, OnePlus) achieved 99.1%–99.6% accuracy (Table 6.9).

7.2.2 RQ2: System Overhead

What is the quantitative computational cost of routing 100% of a device's network traffic through a Kotlin-based interception, classification, and relay pipeline?

Answer: Moderate and acceptable. The QoS Scheduler introduces:

- **CPU:** 12.4% average utilization during normal use, 24.8% peak during sustained high throughput (Table 6.6).
- **Memory:** 68 MB heap allocation (Table 6.7), reduced from 142 MB after the Flyweight optimization.
- **Battery:** 4.5%/hr additional drain, translating to approximately 1 hour of reduced battery life on a 5000 mAh device (Table 6.8).

These overhead levels are comparable to commercial always-on VPN applications and do not cause perceptible device slowdown during normal use. The overhead is dominated by the TUN read-/write system calls and socket relay I/O, both of which are inherent to the user-space VPN architecture

and cannot be significantly optimized without kernel-level access.

7.2.3 RQ3: Latency Mitigation

To what extent does preemptive, user-space active queue management mitigate the latency spikes and jitter associated with network Bufferbloat?

Answer: Dramatically. The QoS Scheduler reduced mean latency by 84.3% (from 245.3 ms to 38.6 ms) during network congestion (Table 6.5). The P99 tail latency was reduced by 91.1% (from 587.4 ms to 52.1 ms), and jitter was reduced by 91.3% (from 89.2 ms to 7.8 ms).

These results were achieved through the "policing and protocol exploitation" design philosophy: by dropping low-priority packets before they reach the hardware modem buffer, the system triggers TCP's congestion control to reduce the remote server's sending rate, indirectly preventing the modem buffer from filling. This mechanism is effective against both CUBIC and BBR congestion control algorithms, though BBR requires approximately 15% more packet drops before converging (Section 6.7).

The 38.6 ms mean latency achieved with QoS-enabled is well within the acceptable threshold for real-time applications (VoIP requires $<$150 ms one-way delay, gaming requires $<$100 ms).

7.3 Limitations

Despite the positive results, the current implementation has several limitations that should be acknowledged:

7.3.1 Technical Limitations

- **DPI with Encrypted Traffic:** The DPI classifier relies primarily on port-based heuristics because TLS 1.3 encryption (and the emerging Encrypted Client Hello standard) prevents payload inspection. Approximately 95% of modern web traffic is encrypted, limiting the classifier to coarse-grained categorization. This limitation is shared by all user-space DPI systems.
- **Uplink Bandwidth Estimation:** The system currently uses a static uplink bandwidth estimate (configured at 100 Mbps by default). In reality, cellular uplink bandwidth varies dynamically with signal strength, network load, and cell handoffs. Without carrier cooperation or kernel-level access to the modem's PHY layer, accurate real-time uplink estimation is extremely challenging.
- **Inbound UDP Limitation:** While the system can drop inbound UDP packets to free local CPU and memory, it cannot reduce the actual bandwidth consumed on the wireless medium. A remote UDP sender will continue transmitting at full speed regardless of local drops, because UDP has no congestion control feedback mechanism.

- **Single VPN Slot:** Android allows only one active VPN connection. The QoS Scheduler cannot coexist with other VPN applications (e.g., corporate VPNs, privacy VPNs). This is a fundamental platform limitation.
- **PacketComposer Fragility:** Manual IP/TCP packet construction is inherently error-prone. Any bug in checksum calculation or header field assignment causes silent packet drops with no error diagnostics, making debugging extremely difficult.

7.3.2 Methodological Limitations

- **Lab Environment:** All system-level tests were conducted in a controlled lab environment with stable Wi-Fi and cellular connectivity. Real-world conditions (variable signal strength, network congestion, mobility) may produce different results.
- **Device Coverage:** Testing covered four devices from four OEMs, representing the major Android ecosystem segments. However, there are hundreds of Android device models with varying hardware capabilities and OEM modifications.
- **Application Diversity:** The test workload consisted primarily of iPerf3 synthetic traffic. Real-world application behavior (multiplayer gaming, video conferencing, web browsing) produces more complex, bursty traffic patterns that may exhibit different scheduling characteristics.

7.4 Future Work

Based on the findings and limitations identified in this thesis, the following research directions are proposed:

7.4.1 Short-Term (6–12 Months)

- **Machine Learning Traffic Classification:** Replace the heuristic-based DPI classifier with a machine learning model trained on flow-level features (packet sizes, inter-arrival times, burst patterns) as proposed by Jiang et al. [31]. This would enable accurate classification of encrypted TLS 1.3 traffic without payload inspection.
- **Adaptive Bandwidth Estimation:** Implement RTT-based uplink bandwidth probing, where the system periodically measures round-trip times to estimate available bandwidth. This would replace the static bandwidth estimate with a dynamic measurement, similar to TCP BBR's bandwidth probing mechanism [12].
- **QUIC-Aware Scheduling:** Extend the scheduler to recognize QUIC traffic (UDP port 443) and apply QUIC-specific policies, leveraging QUIC's connection ID for flow tracking (since QUIC connections survive IP address changes during mobility).

- **IPv6 Full Compliance:** Complete IPv6 extension header chain traversal for all known extension types, including Hop-by-Hop Options, Routing, Fragment, and Destination Options headers.

7.4.2 Medium-Term (1–2 Years)

- **Multi-Device Fleet Synchronization:** Extend the Cloud Management architecture to support fleet-wide QoS policy deployment, with role-based access control, audit logging, and policy versioning.
- **eBPF Integration for Acceleration:** As Android evolves and potentially opens eBPF APIs to applications, integrate kernel-level eBPF programs for performance-critical packet processing while maintaining the user-space control plane. This hybrid architecture could reduce CPU overhead from 12.4% to below 3%.
- **Enterprise MDM Platform Integration:** Develop integration modules for major MDM platforms (Microsoft Intune, VMware Workspace ONE) to add QoS capabilities to existing enterprise device management workflows.
- **User Experience Metrics:** Extend the evaluation methodology to include user-perceived quality metrics: Mean Opinion Score (MOS) for VoIP, video startup time and rebuffering rate for streaming, and frame-time consistency for gaming.

7.4.3 Long-Term (3–5 Years)

- **Edge Computing QoS Orchestration:** Integrate the mobile QoS agent with edge computing nodes (MEC) to enable end-to-end QoS from the mobile device through the edge network to cloud services.
- **5G Network Slicing Integration:** Leverage 5G network slicing APIs (when available) to coordinate device-level QoS with network-level slicing, providing true end-to-end quality guarantees.
- **AI-Driven Autonomous QoS:** Develop reinforcement learning agents that continuously optimize QoS parameters based on real-time network conditions and user behavior patterns, eliminating the need for manual policy configuration.
- **Cross-Platform Expansion:** Port the QoS engine to iOS (using NEPacketTunnelProvider) and desktop platforms (using WFP on Windows, Network Extensions on macOS) to provide a unified, cross-platform QoS solution.

7.5 Lessons Learned

7.5.1 Technical Lessons

- **Lock-free concurrency is essential in the hot path.** The initial prototype used Kotlin `Mutex` for thread safety, which introduced contention under high packet rates. Switching to `ConcurrentHashMap`, `AtomicLong`, and `@Synchronized` methods eliminated the bottleneck without sacrificing safety.
- **Protocol nature is more powerful than brute-force control.** Rather than attempting to buffer and delay packets (shaping), leveraging TCP's built-in congestion control via packet drops (policing) provided superior results with dramatically less implementation complexity.
- **Caching transforms performance at scale.** The five-tier caching architecture reduced the per-packet processing cost from $O(n)$ (with system calls) to $O(1)$ (with cache hits), enabling the system to process thousands of packets per second on a mobile device.
- **Silent failures are the hardest bugs.** The PacketComposer's checksum bugs produced zero error messages—the only symptom was "no Internet." Extensive unit tests with known-good packet captures were essential for validating correctness.

7.5.2 Methodological Lessons

- **Reproducibility requires extreme control.** Network measurements are notoriously variable. The decision to use airplane mode between trials, force-stop background apps, and conduct 10 repetitions with statistical analysis was critical for producing reliable, publishable results.
- **OEM fragmentation is a first-class concern.** The Xiaomi `SecurityException` bug would have been undetectable if testing had been limited to a single device. Cross-OEM testing must be a mandatory part of any Android system-level application validation.

7.6 Broader Impact

7.6.1 Societal Impact

The QoS Scheduler has the potential to democratize network quality management. Currently, QoS tools are exclusively available to network administrators with root access or specialized infrastructure. By operating on stock, unrooted devices, this system enables:

- **Digital equity:** Students in bandwidth-constrained environments can prioritize educational applications over entertainment, ensuring access to learning resources.
- **Parental controls:** Parents can manage children's bandwidth consumption without expensive third-party services or device rooting.

- **Small business optimization:** Small businesses sharing a single Internet connection can ensure that point-of-sale and communication systems receive priority bandwidth.

7.6.2 Environmental Considerations

By reducing unnecessary bandwidth consumption (dropping low-priority packets before they reach the modem), the QoS Scheduler indirectly reduces the energy consumed by network infrastructure. While the per-device impact is small, fleet-wide deployment across thousands of devices could yield measurable reductions in data center and cellular tower energy consumption.

7.6.3 Privacy and Ethics

The VPN architecture inherently gives the QoS Scheduler access to all network traffic metadata. This raises important privacy considerations:

- **Local-only processing:** All QoS scheduling decisions are made locally on the device. The server receives only aggregated per-application statistics, never individual packet metadata or payload content.
- **Transparency:** The open-source nature of the implementation allows independent security auditing.
- **Informed consent:** Android requires explicit user consent (a system dialog) before activating any VPN, ensuring users are aware that their traffic is being processed.

7.7 Final Remarks

This thesis has demonstrated that the "Root Privilege Wall"—the traditional barrier between consumers and enterprise-grade network QoS tools—can be effectively bypassed through creative use of the Android VpnService API. By combining bit-level packet parsing, nanosecond-precision scheduling algorithms, lock-free concurrent programming, and cloud-based policy management, the QoS Scheduler brings professional-grade network quality management to every Android user.

The system achieves 99.4% rate enforcement accuracy, near-perfect proportional fairness, and 84.3% latency reduction during Bufferbloat conditions—all without requiring root access, kernel modifications, or network infrastructure changes. As mobile devices continue to become the primary computing platform for billions of users worldwide, the ability to intelligently manage network quality at the device edge will become increasingly critical.

The future of mobile QoS lies not in more powerful hardware or larger buffers, but in smarter, user-space software that understands application behavior and allocates resources accordingly. This thesis provides both the theoretical foundation and the practical proof-of-concept for that future.

BIBLIOGRAPHY

[1] Ericsson, "Ericsson mobility report," 2024. Accessed: 2026.

[2] J. Gettys and K. Nichols, "Bufferbloat: Dark buffers in the internet," *Communications of the ACM*, vol. 55, no. 1, pp. 57–65, 2012.

[3] K. Nichols, S. Blake, F. Baker, and D. Black, "Definition of the differentiated services field (ds field) in the ipv4 and ipv6 headers," RFC 2474, IETF, 1998.

[4] A. Demers, S. Keshav, and S. Shenker, "Analysis and simulation of a fair queueing algorithm," *ACM SIGCOMM Computer Communication Review*, vol. 19, no. 4, pp. 1–12, 1989.

[5] A. S. Tanenbaum and D. J. Wetherall, *Computer Networks*. Pearson, 5th ed., 2011.

[6] J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach*. Pearson, 8th ed., 2021.

[7] J. Postel, "Internet protocol," RFC 791, IETF, 1981.

[8] S. Deering and R. Hinden, "Internet protocol, version 6 (ipv6) specification," RFC 8200, IETF, 2017.

[9] W. Eddy, "Transmission control protocol (tcp)," RFC 9293, IETF, 2022.

[10] M. Allman, V. Paxson, and E. Blanton, "Tcp congestion control," RFC 5681, IETF, 2009.

[11] S. Ha, I. Rhee, and L. Xu, "Cubic: A new tcp-friendly high-speed tcp variant," *ACM SIGOPS Operating Systems Review*, vol. 42, no. 5, pp. 64–74, 2008.

[12] N. Cardwell, Y. Cheng, C. S. Gunn, S. H. Yeganeh, and V. Jacobson, "Bbr: Congestion-based congestion control," *ACM Queue*, vol. 14, no. 5, pp. 20–53, 2017.

[13] J. Postel, "User datagram protocol," RFC 768, IETF, 1980.

[14] J. Iyengar and M. Thomson, "Quic: A udp-based multiplexed and secure transport," RFC 9000, IETF, 2021.

[15] W. Stallings, *Data and Computer Communications*. Pearson, 10th ed., 2014.

[16] S. Floyd and V. Jacobson, "Random early detection gateways for congestion avoidance," *IEEE/ACM Transactions on Networking*, vol. 1, no. 4, pp. 397–413, 1993.

[17] K. Nichols, V. Jacobson, A. McGregor, and J. Iyengar, "Controlled delay active queue management," RFC 8289, IETF, 2018.

[18] T. Hoeiland-Joergensen, P. McKenney, D. Taht, J. Gettys, and E. Dumazet, “The flow queue codel packet scheduler and active queue management algorithm,” RFC 8290, IETF, 2018.

[19] J. Heinanen and R. Guerin, “A single rate three color marker,” RFC 2697, IETF, 1999.

[20] J. Heinanen and R. Guerin, “A two rate three color marker,” RFC 2698, IETF, 1999.

[21] A. K. Parekh and R. G. Gallager, “A generalized processor sharing approach to flow control in integrated services networks: The single-node case,” *IEEE/ACM Transactions on Networking*, vol. 1, no. 3, pp. 344–357, 1993.

[22] Google, “Vpnservice – android developers,” 2024. Accessed: 2026.

[23] M. A. M. Vieira, M. S. Castanho, R. D. G. Pacífico, E. R. S. Santos, E. P. M. C. Júnior, and L. F. M. Coelho, “Fast packet processing with ebpf and xdp: Concepts, code, challenges, and applications,” in *ACM Computing Surveys*, vol. 53, pp. 1–36, ACM, 2020.

[24] M. Bokhorst, “Netguard – no-root firewall,” 2023. GitHub Repository.

[25] AFWall+ Team, “Afwall+ – android firewall+,” 2023. GitHub Repository.

[26] GlassWire, “Glasswire – data usage monitor,” 2024. Accessed: 2026.

[27] M. S. Seddiki, M. Shahbaz, S. Donovan, S. Grover, M. Park, N. Feamster, and Y.-Q. Song, “Flowqos: Qos for the rest of us,” in *ACM Workshop on Hot Topics in Networks (HotNets)*, pp. 1–7, ACM, 2015.

[28] Gartner, “Magic quadrant for unified endpoint management tools,” 2024. Accessed: 2026.

[29] R. Braden, D. Borman, and C. Partridge, “Computing the internet checksum,” RFC 1071, IETF, 1988.

[30] J.-Y. Le Boudec and P. Thiran, *Network Calculus: A Theory of Deterministic Queuing Systems for the Internet*, vol. 2050 of *Lecture Notes in Computer Science*. Springer, 2001.

[31] Y. Jiang *et al.*, “Understanding android vpn ecosystem in the wild,” in *ACM Internet Measurement Conference (IMC)*, pp. 1–14, ACM, 2021.

APPENDIX A

KEY SOURCE CODE MODULES

This appendix presents the complete source code of the most critical modules in the QoS Scheduler system. These modules form the core of the Data Plane pipeline and are referenced throughout Chapters 4 and 5.

> **Note to the Evaluation Committee:** The full, unedited source code of the entire Android application and Node.js management server is available open-source on GitHub. Please scan the QR code below or visit the URL to access the repository: > > **GitHub Repository:** https://github.com/hieum/qos-scheduler-android > > *(A printed QR code pointing to the URL above should be inserted here before final binding)*

A.1 Packet Parser — `RawPacket.kt`

The `RawPacket` module is responsible for parsing raw byte arrays read from the TUN interface into structured packet objects. It supports both IPv4 and IPv6 with full Extension Header Chaining.

```
package com.qos.scheduler.model

import java.nio.ByteBuffer

data class RawPacket(
    val ipVersion: Int,
    val srcIp: String, val dstIp: String,
    val srcPort: Int, val dstPort: Int,
    val protocol: Protocol,
    val length: Int,
    val ipHeaderLength: Int,
    val transportHeaderLength: Int,
    val payloadOffset: Int,
    val payloadLength: Int,
    val tcpSequenceNumber: Long? = null,
    val tcpAckNumber: Long? = null,
    val tcpWindowSize: Int? = null,
    val tcpFlags: TcpControlFlags? = null,
    val rawBuffer: ByteArray
) {
    val isIpv4: Boolean get() = ipVersion == 4
    val payload: ByteArray
```

```
        get() = rawBuffer.copyOfRange(
            payloadOffset, payloadOffset + payloadLength)

    companion object {
        fun parse(buffer: ByteArray, length: Int): RawPacket? {
            if (length < 20) return null
            val buf = ByteBuffer.wrap(buffer, 0, length)
            val version = (buf.get(0).toInt() and 0xFF) shr 4
            return when (version) {
                4 -> parseIpv4(buf, buffer, length)
                6 -> parseIpv6(buf, buffer, length)
                else -> null
            }
        }

        private fun parseIpv4(
            buf: ByteBuffer, raw: ByteArray, length: Int
        ): RawPacket? {
            val ihl = (buf.get(0).toInt() and 0x0F) * 4
            if (length < ihl + 4) return null
            val protocolNum = buf.get(9).toInt() and 0xFF
            val protocol = Protocol.fromNumber(protocolNum)
            val srcIp = buildString {
                for (i in 12..15) {
                    if (i > 12) append('.')
                    append(buf.get(i).toInt() and 0xFF)
                }
            }
            val dstIp = buildString {
                for (i in 16..19) {
                    if (i > 16) append('.')
                    append(buf.get(i).toInt() and 0xFF)
                }
            }
            val (srcPort, dstPort) = extractPorts(
                buf, ihl, protocol)
            val tcpMeta = extractTcpMetadata(
                buf, ihl, protocol, length)
            val thl = when (protocol) {
                Protocol.TCP -> tcpMeta.headerLength
                Protocol.UDP -> 8
                else -> 0
            }
            val po = (ihl + thl).coerceAtMost(length)
            val pl = (length - po).coerceAtLeast(0)
            return RawPacket(
```

```
                ipVersion = 4, srcIp = srcIp,
                dstIp = dstIp, srcPort = srcPort,
                dstPort = dstPort, protocol = protocol,
                length = length, ipHeaderLength = ihl,
                transportHeaderLength = thl,
                payloadOffset = po, payloadLength = pl,
                tcpSequenceNumber = tcpMeta.sequenceNumber,
                tcpAckNumber = tcpMeta.ackNumber,
                tcpWindowSize = tcpMeta.windowSize,
                tcpFlags = tcpMeta.flags,
                rawBuffer = raw.copyOf(length)
            )
        }

        private fun parseIpv6(
            buf: ByteBuffer, raw: ByteArray, length: Int
        ): RawPacket? {
            if (length < 40) return null
            var nextHeader = buf.get(6).toInt() and 0xFF
            var headerOffset = 40
            while (isIpv6ExtensionHeader(nextHeader)
                && headerOffset < length) {
                if (headerOffset + 2 > length) return null
                val extNext =
                    buf.get(headerOffset).toInt() and 0xFF
                val extLength = when (nextHeader) {
                    44 -> 8 // Fragment: fixed 8 bytes
                    else -> (buf.get(headerOffset + 1)
                        .toInt() and 0xFF) * 8 + 8
                }
                nextHeader = extNext
                headerOffset += extLength
            }
            val protocol = Protocol.fromNumber(nextHeader)
            val srcIp = formatIpv6(buf, 8)
            val dstIp = formatIpv6(buf, 24)
            val (srcPort, dstPort) = extractPorts(
                buf, headerOffset, protocol)
            val thl = if (protocol == Protocol.TCP
                && headerOffset + 13 < length) {
                ((buf.get(headerOffset + 12).toInt()
                    ushr 4) and 0x0F) * 4
            } else 8
            val po = (headerOffset+thl).coerceAtMost(length)
            val pl = (length - po).coerceAtLeast(0)
            return RawPacket(
```

```
                ipVersion = 6, srcIp = srcIp,
                dstIp = dstIp, srcPort = srcPort,
                dstPort = dstPort, protocol = protocol,
                length = length,
                ipHeaderLength = headerOffset,
                transportHeaderLength = thl,
                payloadOffset = po, payloadLength = pl,
                rawBuffer = raw.copyOf(length)
            )
        }

        private fun isIpv6ExtensionHeader(nh: Int) =
            nh in setOf(0, 43, 44, 60)

        private fun extractPorts(
            buf: ByteBuffer, off: Int, proto: Protocol
        ): Pair<Int, Int> {
            if (proto == Protocol.OTHER) return Pair(0, 0)
            if (buf.capacity() < off + 4) return Pair(0, 0)
            val src = ((buf.get(off).toInt() and 0xFF) shl 8
                ) or (buf.get(off+1).toInt() and 0xFF)
            val dst = ((buf.get(off+2).toInt() and 0xFF
                ) shl 8) or (buf.get(off+3).toInt() and 0xFF)
            return Pair(src, dst)
        }

        private fun extractTcpMetadata(
            buf: ByteBuffer, off: Int,
            proto: Protocol, len: Int
        ): TcpMetadata {
            if (proto != Protocol.TCP || off+20 > len)
                return TcpMetadata(0, null, null, null, null)
            val seq = buf.getInt(off+4).toLong() and
                0xFFFFFFFFL
            val ack = buf.getInt(off+8).toLong() and
                0xFFFFFFFFL
            val dataOff = ((buf.get(off+12).toInt()
                ushr 4) and 0x0F) * 4
            val flagsByte = buf.get(off+13).toInt() and 0xFF
            val win = buf.getShort(off+14).toInt() and 0xFFFF
            val flags = TcpControlFlags(
                fin = (flagsByte and 0x01) != 0,
                syn = (flagsByte and 0x02) != 0,
                rst = (flagsByte and 0x04) != 0,
                psh = (flagsByte and 0x08) != 0,
                ack = (flagsByte and 0x10) != 0
```

```
            )
            return TcpMetadata(dataOff, seq, ack, win, flags)
        }

        // IPv6 address formatting with :: compression
        private fun formatIpv6(
            buf: ByteBuffer, offset: Int
        ): String {
            val segs = IntArray(8) { i ->
                val h = buf.get(offset+i*2).toInt() and 0xFF
                val l = buf.get(offset+i*2+1).toInt() and 0xFF
                (h shl 8) or l
            }
            var maxStart = -1; var maxLen = 0
            var curStart = -1; var curLen = 0
            for (i in segs.indices) {
                if (segs[i] == 0) {
                    if (curStart == -1) {
                        curStart = i; curLen = 1
                    } else curLen++
                } else {
                    if (curLen > maxLen) {
                        maxStart = curStart
                        maxLen = curLen
                    }
                    curStart = -1; curLen = 0
                }
            }
            if (curLen > maxLen) {
                maxStart = curStart; maxLen = curLen
            }
            return buildString {
                var i = 0
                var justEmitted = false
                while (i < 8) {
                    if (i == maxStart && maxLen > 1) {
                        append("::"); i += maxLen
                        justEmitted = true
                        if (i >= 8) break
                    } else {
                        if (i > 0 && !justEmitted) append(':')
                        justEmitted = false
                        append(String.format("%x", segs[i]))
                        i++
                    }
                }
```

```
207                 }
208             }
209
210             private data class TcpMetadata(
211                 val headerLength: Int,
212                 val sequenceNumber: Long?,
213                 val ackNumber: Long?,
214                 val windowSize: Int?,
215                 val flags: TcpControlFlags?
216             )
217         }
218 }
219
220 data class TcpControlFlags(
221     val fin: Boolean = false,
222     val syn: Boolean = false,
223     val rst: Boolean = false,
224     val psh: Boolean = false,
225     val ack: Boolean = false
226 )
```

Listing A.1: RawPacket.kt — Complete packet parser

A.2 Token Bucket — `TokenBucket.kt`

The Token Bucket module implements the rate-limiting algorithm with nanosecond-precision timing. Thread safety is ensured via the `@Synchronized` annotation.

```
1  package com.qos.scheduler.scheduler
2
3  class TokenBucket(
4      @Volatile var rateBps: Long,
5      @Volatile var burstBytes: Long
6  ) {
7      private var tokens: Double = burstBytes.toDouble()
8      private var lastRefillTime: Long = System.nanoTime()
9
10     @Synchronized
11     fun consume(bytes: Int): Boolean {
12         refill()
13         return if (tokens >= bytes) {
14             tokens -= bytes
15             true
16         } else {
17             false
```

```
        }
    }

    @Synchronized
    fun setRate(newRateBps: Long) {
        rateBps = newRateBps
        burstBytes = newRateBps * 2
        tokens = minOf(tokens, burstBytes.toDouble())
    }

    private fun refill() {
        val now = System.nanoTime()
        val elapsed =
            (now - lastRefillTime) / 1_000_000_000.0
        tokens = minOf(burstBytes.toDouble(),
                       tokens + rateBps * elapsed)
        lastRefillTime = now
    }
}
```

Listing A.2: TokenBucket.kt — Complete rate limiter

A.3 Bandwidth Scheduler — `BandwidthScheduler.kt`

The Bandwidth Scheduler orchestrates all Token Buckets and implements the Weighted Fair Queuing rebalancing algorithm.

```
package com.qos.scheduler.scheduler

import com.qos.scheduler.model.AppTraffic
import com.qos.scheduler.model.TrafficClass
import java.util.concurrent.ConcurrentHashMap

class BandwidthScheduler {
    var uplinkBps: Long = 50_000_000L
    private val buckets =
        ConcurrentHashMap<String, TokenBucket>()
    private val manualCaps =
        ConcurrentHashMap<String, Long>()

    companion object {
        fun appBucketKey(uid: Int) = "app_$uid"
    }

    fun processPacket(key: String, size: Int): Boolean {
```

```
        val bucket = buckets[key] ?: return true
        return bucket.consume(size)
    }

    fun ensureBucket(
        key: String, priority: TrafficClass = TrafficClass.MEDIUM
    ) {
        if (!buckets.containsKey(key)) {
            val (rate, burst) = rateForClass(priority)
            buckets[key] = TokenBucket(rate, burst)
        }
    }

    fun setManualCap(ip: String, rateBps: Long) {
        manualCaps[ip] = rateBps
        buckets[ip]?.setRate(rateBps)
    }

    fun rebalanceWithApps(apps: Collection<AppTraffic>) {
        val hc = apps.count {
            it.priorityClass == TrafficClass.HIGH }
        val mc = apps.count {
            it.priorityClass == TrafficClass.MEDIUM }
        val lc = apps.count {
            it.priorityClass == TrafficClass.LOW }
        val totalWeight = hc*4 + mc*2 + lc*1 + 2
        if (totalWeight <= 0) return

        apps.forEach { app ->
            val key = appBucketKey(app.uid)
            if (manualCaps.containsKey(key)) return@forEach
            val weight = when (app.priorityClass) {
                TrafficClass.HIGH -> 4
                TrafficClass.MEDIUM -> 2
                TrafficClass.LOW -> 1
            }
            val rate = (uplinkBps * weight) / totalWeight
            buckets.getOrPut(key) {
                TokenBucket(rate, rate * 5) }.setRate(rate)
        }

        val hostRate = (uplinkBps * 2) / totalWeight
        buckets.getOrPut("__host__") {
            TokenBucket(hostRate, hostRate * 5)
        }.setRate(hostRate)
    }
```

```
65
66      private fun rateForClass(
67          cls: TrafficClass
68      ): Pair<Long, Long> {
69          val rate = when (cls) {
70              TrafficClass.HIGH -> (uplinkBps * 0.8).toLong()
71              TrafficClass.MEDIUM -> (uplinkBps * 0.5).toLong()
72              TrafficClass.LOW -> (uplinkBps * 0.2).toLong()
73          }
74          val burst = when (cls) {
75              TrafficClass.HIGH -> rate * 2
76              TrafficClass.MEDIUM -> rate
77              TrafficClass.LOW -> rate / 2
78          }
79          return Pair(rate, burst)
80      }
81  }
```

Listing A.3: BandwidthScheduler.kt — WFQ orchestrator

A.4 DPI Classifier — `DpiClassifier.kt`

The DPI-Lite classifier categorizes traffic based on destination port heuristics.

```
1  package com.qos.scheduler.classifier
2
3  import com.qos.scheduler.model.Protocol
4  import com.qos.scheduler.model.RawPacket
5  import com.qos.scheduler.model.TrafficCategory
6
7  class DpiClassifier {
8      fun classify(packet: RawPacket): TrafficCategory {
9          val port = packet.dstPort
10         val protocol = packet.protocol
11
12         // VoIP / Real-time (High Priority)
13         if (protocol == Protocol.UDP) {
14             if (port in 5060..5061
15                 || port in 10000..20000)
16                 return TrafficCategory.VOIP
17         }
18
19         // Gaming (High Priority)
20         val gamingPorts = listOf(
21             27015, 27016, 5000, 5500, 8001, 8002)
```

```
        if (gamingPorts.contains(port))
            return TrafficCategory.ONLINE_GAMING

        // DNS
        if (port == 53)
            return TrafficCategory.WEB_BROWSING

        // Video Streaming
        if (port == 1935 || port == 8080)
            return TrafficCategory.STREAMING

        // Web / Social
        if (port == 80 || port == 443)
            return TrafficCategory.WEB_BROWSING

        // Downloads (Low Priority)
        val dlPorts = listOf(21, 22, 143, 993, 110, 995)
        if (dlPorts.contains(port))
            return TrafficCategory.FILE_TRANSFER

        return TrafficCategory.UNKNOWN
    }
}
```

Listing A.4: DpiClassifier.kt — Port-based traffic classifier

A.5 Data Plane Processor — `DataPlaneProcessor.kt`

The Data Plane Processor orchestrates the entire per-packet pipeline: parsing, UID resolution, classification, and scheduling decisions.

```
package com.qos.scheduler.service.dataplane

import com.qos.scheduler.classifier.DpiClassifier
import com.qos.scheduler.model.*
import com.qos.scheduler.scheduler.BandwidthScheduler
import com.qos.scheduler.util.AppResolver
import java.util.concurrent.ConcurrentHashMap

class DataPlaneProcessor(
    private val classifier: DpiClassifier,
    private val scheduler: BandwidthScheduler,
    private val appResolver: AppResolver,
    private val appTraffic:
        ConcurrentHashMap<Int, AppTraffic>,
```

```
    private val hostBucketKey: String
) {
    companion object {
        private val flowCache =
            ConcurrentHashMap<FlowKey, Int>()
    }

    data class Result(
        val parsed: Boolean,
        val droppedByScheduler: Boolean,
        val qosKey: String,
        val packet: RawPacket? = null
    )

    fun process(
        bytes: ByteArray, length: Int, isOutbound: Boolean
    ): Result {
        val pkt = RawPacket.parse(bytes, length)
            ?: return Result(false, false, hostBucketKey)

        val flowKey = if (isOutbound)
            FlowKey(pkt.srcIp, pkt.dstIp,
                pkt.srcPort, pkt.dstPort, pkt.protocol)
        else
            FlowKey(pkt.dstIp, pkt.srcIp,
                pkt.dstPort, pkt.srcPort, pkt.protocol)

        // STEP 1: UID Resolution (~1ms syscall)
        var uid = flowCache[flowKey]
        if (uid == null) {
            val pNum = if (pkt.protocol == Protocol.TCP)
                6 else 17
            val resolved = if (isOutbound)
                appResolver.getUidForConnection(pNum,
                    pkt.srcIp, pkt.srcPort,
                    pkt.dstIp, pkt.dstPort)
            else appResolver.getUidForConnection(pNum,
                    pkt.dstIp, pkt.dstPort,
                    pkt.srcIp, pkt.srcPort)

            if (resolved != null && resolved > 0) {
                flowCache[flowKey] = resolved
                uid = resolved
                // Retroactive migration of synthetic
                // port buckets occurs here (omitted
                // for brevity)
```

```
            }
        }

        // STEP 2: Classify & Track
        val cat = classifier.classify(pkt)
        var sKey = hostBucketKey
        if (uid != null && uid > 0) {
            val app = appTraffic.getOrPut(uid) {
                val info = appResolver.getAppInfo(uid)
                AppTraffic(uid, info.appName,
                    info.packageName, TrafficClass.MEDIUM)
            }
            sKey = BandwidthScheduler.appBucketKey(uid)
            if (isOutbound) app.bytesOut += length
            else app.bytesIn += length
            scheduler.ensureBucket(sKey, app.priorityClass)
        }

        // STEP 3: Scheduler Decision
        val allowed = if (isOutbound)
            scheduler.processPacket(sKey, length) else true

        return Result(true, !allowed, sKey, pkt)
    }
}
```

Listing A.5: DataPlaneProcessor.kt — Pipeline orchestrator

APPENDIX B

PACKET COMPOSER AND USER GUIDE

B.1 Packet Composer — `PacketComposer.kt`

The `PacketComposer` module synthesizes valid IP packets for injection back into the TUN interface. It implements the RFC 1071 Internet Checksum algorithm with pseudo-header support for both IPv4 and IPv6.

```
object PacketComposer {

    fun composeTcpV4(
        srcIp: String, srcPort: Int,
        dstIp: String, dstPort: Int,
        sequenceNumber: Long, ackNumber: Long,
        flags: TcpControlFlags,
        windowSize: Int = 65535,
        payload: ByteArray = byteArrayOf()
    ): ByteArray {
        val tcpHL = 20
        val totalLength = 20 + tcpHL + payload.size
        val buf = ByteBuffer.allocate(totalLength)

        // --- IPv4 Header (20 bytes) ---
        buf.put(0, (0x45).toByte()) // Ver=4, IHL=5
        buf.put(1, 0)               // DSCP + ECN
        buf.putShort(2, totalLength.toShort())
        buf.putShort(4, (Math.random() * 65535)
            .toInt().toShort())     // Random ID
        buf.putShort(6, 0)          // Flags + Frag
        buf.put(8, 64)              // TTL = 64
        buf.put(9, 6)               // Protocol = TCP
        buf.putShort(10, 0)         // Checksum placeholder

        val srcBytes = ipToBytes(srcIp)
        val dstBytes = ipToBytes(dstIp)
        buf.position(12); buf.put(srcBytes)
        buf.put(dstBytes)
        buf.putShort(10, computeIpChecksum(buf, 0, 20))

```

```
32          // --- TCP Header (20 bytes) ---
33          val tcpStart = 20
34          buf.position(tcpStart)
35          buf.putShort(srcPort.toShort())
36          buf.putShort(dstPort.toShort())
37          buf.putInt(sequenceNumber.toInt())
38          buf.putInt(ackNumber.toInt())
39
40          var flagsBits = (tcpHL / 4) shl 12
41          if (flags.fin) flagsBits = flagsBits or 0x01
42          if (flags.syn) flagsBits = flagsBits or 0x02
43          if (flags.rst) flagsBits = flagsBits or 0x04
44          if (flags.psh) flagsBits = flagsBits or 0x08
45          if (flags.ack) flagsBits = flagsBits or 0x10
46          buf.putShort(flagsBits.toShort())
47          buf.putShort(windowSize.toShort())
48          buf.putShort(0) // TCP Checksum placeholder
49          buf.putShort(0) // Urgent pointer
50
51          if (payload.isNotEmpty()) {
52              buf.position(tcpStart + tcpHL)
53              buf.put(payload)
54          }
55
56          // --- TCP Checksum with Pseudo-header ---
57          val chk = computeTcpChecksum(buf, tcpStart,
58              tcpHL + payload.size, srcBytes, dstBytes)
59          buf.putShort(tcpStart + 16, chk)
60
61          return buf.array()
62      }
63  }
```

Listing B.1: PacketComposer.kt — IPv4 TCP packet synthesis

```
1  // IP Header Checksum
2  private fun computeIpChecksum(
3      buffer: ByteBuffer, offset: Int, length: Int
4  ): Short {
5      var sum = 0L
6      for (i in 0 until length / 2) {
7          sum += (buffer.getShort(offset + i * 2)
8              .toInt() and 0xFFFF).toLong()
9      }
10     while (sum shr 16 != 0L) {
11         sum = (sum and 0xFFFF) + (sum shr 16)
```

```
    }
    return (sum.inv() and 0xFFFF).toShort()
}

// TCP Checksum with Pseudo-header
private fun computeTcpChecksum(
    buffer: ByteBuffer, tcpOffset: Int,
    tcpLength: Int,
    srcIp: ByteArray, dstIp: ByteArray
): Short {
    var sum = 0L
    // Pseudo-header: src IP + dst IP
    if (srcIp.size == 4) {
        sum += ((srcIp[0].toInt() and 0xFF) shl 8
            or (srcIp[1].toInt() and 0xFF)).toLong()
        sum += ((srcIp[2].toInt() and 0xFF) shl 8
            or (srcIp[3].toInt() and 0xFF)).toLong()
        sum += ((dstIp[0].toInt() and 0xFF) shl 8
            or (dstIp[1].toInt() and 0xFF)).toLong()
        sum += ((dstIp[2].toInt() and 0xFF) shl 8
            or (dstIp[3].toInt() and 0xFF)).toLong()
    }
    sum += 6L            // Protocol: TCP
    sum += tcpLength.toLong()

    // TCP Header + Payload
    for (i in 0 until tcpLength / 2) {
        sum += (buffer.getShort(tcpOffset + i * 2)
            .toInt() and 0xFFFF).toLong()
    }
    if (tcpLength % 2 != 0) {
        sum += ((buffer.get(tcpOffset + tcpLength - 1)
            .toInt() and 0xFF) shl 8).toLong()
    }
    // Fold carries
    while (sum shr 16 != 0L) {
        sum = (sum and 0xFFFF) + (sum shr 16)
    }
    val result = (sum.inv() and 0xFFFF).toShort()
    return if (result == (0).toShort())
        (0xFFFF).toShort() else result
}
```

Listing B.2: Internet Checksum (RFC 1071) implementation

B.2 Application User Guide

This section provides a step-by-step guide for installing and using the QoS Scheduler application.

B.2.1 System Requirements

- Android 10 (API Level 29) or higher.
- No Root access required.
- At least 100 MB of free storage.
- Active WiFi or mobile data connection.

B.2.2 Installation

1. Open the project in Android Studio.
2. Connect the target Android device via USB with USB Debugging enabled.
3. Select the device in the toolbar and click **Run** (Shift+F10).
4. Alternatively, generate a signed APK via **Build** → **Generate Signed Bundle / APK** and install it manually on the device.

B.2.3 First Launch

1. Open the **QoS Scheduler** application.
2. The system will request VPN permission: *"QoS Scheduler wants to set up a VPN connection"*. Tap **OK** to grant permission.
3. The Dashboard screen will appear, showing:
    - VPN connection status indicator.
    - Real-time uplink/downlink throughput display.
    - List of detected applications and their traffic statistics.

B.2.4 Configuring QoS Policies

B.2.4.1 Setting Application Priority

1. Tap on any application in the traffic list.

2. Select the desired priority level:
    - **HIGH** (Weight = 4): For latency-sensitive apps (gaming, video calls).
    - **MEDIUM** (Weight = 2): Default for general use (web browsing).
    - **LOW** (Weight = 1): For background tasks (downloads, sync).
3. The change takes effect immediately — no VPN restart required.

B.2.4.2 Setting Manual Bandwidth Caps

1. Tap on an application in the traffic list.
2. Enter a manual bandwidth limit in Mbps (e.g., `1.0` for 1 Mbps).
3. The Token Bucket for that application is immediately reconfigured.
4. To remove the cap, set it to `0` or delete the value.

B.2.4.3 Configuring Total Uplink Bandwidth

1. Navigate to **Settings** via the gear icon.
2. Enter the estimated total uplink bandwidth in Mbps.
3. This value is used as R_{total} in the WFQ formula (Equation **??**) to calculate per-application allocations.

B.2.5 Monitoring

The Dashboard provides real-time information:

- **Per-Application Statistics:** Bytes sent/received, active flow count, current throughput.
- **QoS Metrics:** Allowed packets vs. dropped packets per application, indicating how aggressively the Token Bucket is throttling each app.
- **System Status:** VPN uptime, total processed packets, relay health indicators.

B.2.6 Stopping the Service

1. Tap the **Stop QoS** button on the Dashboard.
2. The VPN tunnel is torn down, the foreground notification is dismissed, and all network traffic returns to the default best-effort routing.
3. Alternatively, the service can be stopped from the Android system VPN settings.

B.2.7 Troubleshooting

Table B.1: Common issues and solutions

Issue	Solution
No Internet after enabling VPN	Ensure the device has an active network connection. Try disabling and re-enabling the VPN. Check that no other VPN application is running simultaneously.
App not detected in traffic list	The app may not have generated any network traffic yet. Open the app and perform a network action (e.g., load a webpage).
Battery drain	QoS processing requires continuous CPU usage. Consider setting the QoS uplink bandwidth to match your actual connection speed to reduce unnecessary processing.
MIUI SecurityException	On Xiaomi devices, go to **Settings** → **Apps** → **QoS Scheduler** → **Permissions** and ensure all network-related permissions are granted.